Chapter-1 Introductio to Electrostatics Problem Solutions

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Problem 1.1

1.1 Problem Statement

Use Gauss's theorem [and Eq. (1.21) if necessary] to prove the following:

(a) Any excess charge placed on a conductor must lie entirely on its surface. (A conductor by definition contains charges capable of moving freely under the action of applied electric fields.)

(b) A closed, hollow conductor shields its interior from fields due to charges outside, but does not shield its exterior from the fields due to charges placed inside it.

(c) The electric field at the surface of a conductor is normal to the surface and has a magnitude σ/ε_0, where σ is the charge density per unit area on the surface.

1.2 Solution

Physical principles

A conductor contains free charges. If an electric field existed inside the conducting material, these charges would feel a force

$$\mathbf{F} = q\mathbf{E},$$

and they would move. But in electrostatic equilibrium no macroscopic motion of charge occurs. Therefore the electric field inside the conductor must vanish:

$$\boxed{\mathbf{E} = 0 \quad \text{inside the conducting material.}}$$

This immediately implies that the potential is constant throughout the conductor. Indeed, using

$$V(\mathbf{b}) - V(\mathbf{a}) = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l},$$

if $\mathbf{E} = 0$ everywhere inside the conductor, then

$$V(\mathbf{b}) - V(\mathbf{a}) = 0.$$

Hence

$$\boxed{V = \text{constant throughout the conductor.}}$$

(a) Excess charge resides on the surface

Take any closed Gaussian surface S lying entirely within the conducting material. Since the conductor is in electrostatic equilibrium,

$$\mathbf{E} = 0 \quad \text{everywhere on } S.$$

Therefore the flux through S is

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = 0.$$

By Gauss's law,

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\varepsilon_0},$$

so

$$\frac{Q_{\text{enc}}}{\varepsilon_0} = 0 \qquad \Rightarrow \qquad Q_{\text{enc}} = 0.$$

Thus any closed surface taken entirely inside the conductor encloses zero net charge.

Suppose, for contradiction, that some excess charge were located somewhere in the interior of the conductor. Then one could draw a Gaussian surface entirely inside the conductor that encloses that excess charge. But Gauss's law would then require $Q_{\text{enc}} \neq 0$, contradicting the result above. Therefore no excess charge can remain in the bulk of the conductor.

Hence

$$\boxed{\text{Any excess charge placed on a conductor lies entirely on its surface.}}$$

(b) Shielding properties of a closed hollow conductor

First consider a closed hollow conductor with an empty cavity and charges outside the conductor. Since $\mathbf{E} = 0$ in the conducting material, the conductor is at constant potential V_0. In particular, the inner surface of the cavity is also at potential V_0.

Inside the empty cavity, $\rho = 0$, so the potential satisfies Laplace's equation,

$$\nabla^2 V = 0.$$

The boundary condition on the cavity wall is

$$V = V_0 = \text{constant}.$$

A constant function $V(\mathbf{r}) = V_0$ satisfies Laplace's equation and the boundary condition, and by uniqueness this is the only solution. Therefore

$$\mathbf{E} = -\nabla V = 0 \quad \text{inside the cavity.}$$

So the cavity is shielded from outside fields:

$$\boxed{\mathbf{E} = 0 \quad \text{inside the empty cavity.}}$$

Now place a charge q inside the cavity. Draw a closed Gaussian surface lying entirely in the conducting material and surrounding the cavity. Since it lies in the conductor,

$$\mathbf{E} = 0,$$

so Gauss's law gives

$$Q_{\text{enc}} = 0.$$

The enclosed charge consists of the point charge q and the induced charge Q_{inner} on the inner surface, so

$$q + Q_{\text{inner}} = 0,$$

which implies

$$\boxed{Q_{\text{inner}} = -q.}$$

If the conductor as a whole is initially neutral, then the outer surface must carry charge

$$\boxed{Q_{\text{outer}} = +q.}$$

This nonzero outer charge produces an electric field outside the conductor. Therefore the exterior is not shielded from charges inside the cavity.

(c) Field at the surface of a conductor

Suppose there were a tangential component of the electric field at the conductor surface:

$$\mathbf{E}_{\parallel} \neq 0.$$

Then a charge on the surface would feel a tangential force

$$\mathbf{F}_{\parallel} = q\mathbf{E}_{\parallel},$$

so free charges would move along the surface. But in electrostatic equilibrium this cannot happen. Hence

$$\boxed{\mathbf{E}_{\parallel} = 0.}$$

Thus the electric field at the surface must be normal to the surface.

To find its magnitude, take a very small cylindrical pillbox Gaussian surface straddling the conductor surface, with top face area A just outside and bottom face area A just inside. Since the field inside the conductor is zero,

$$\Phi_{\text{bottom}} = 0.$$

The side contribution vanishes in the infinitesimal limit, and the flux through the top face is

$$\Phi_{\text{top}} = E_{\perp}^{\text{out}} A.$$

Hence

$$\oint \mathbf{E} \cdot d\mathbf{a} = E_{\perp}^{\text{out}} A.$$

The enclosed charge is the surface charge in the pillbox:

$$Q_{\text{enc}} = \sigma A.$$

Applying Gauss's law,

$$E_{\perp}^{\text{out}} A = \frac{\sigma A}{\varepsilon_0}.$$

Canceling A gives

$$E_{\perp}^{\text{out}} = \frac{\sigma}{\varepsilon_0}.$$

Therefore

$$\boxed{\mathbf{E}_{\text{outside}} = \frac{\sigma}{\varepsilon_0} \hat{\mathbf{n}}.}$$

Problem 1.24

2.1 Problem Statement

Consider solution of the two-dimensional Poisson equation problem of Problem 1.21, a unit square with zero potential on the boundary and a constant unit charge density in the interior, by the technique of relaxation. Choose $h = 0.25$ so that there are nine interior sites. Use symmetry to reduce the number of needed sites to three, at $(0.25, 0.25)$, $(0.5, 0.25)$, and $(0.5, 0.5)$. With so few sites, it is easy to do the iterations with a block of paper and a pocket calculator, but suit yourself.

(a) Use the "improved grid" averaging of Problem 1.22 and the simple (Jacobian) iteration scheme, starting with $4\pi\varepsilon_0\Phi = 1.0$ at all three interior sites. Do at least six iterations, preferably eight or ten.

(b) Repeat the iteration procedure with the same starting values, but using Gauss–Seidel iteration.

(c) Graph the two sets of results of each iteration versus iteration number and compare with the exact values, $4\pi\varepsilon_0\Phi(0.25, 0.25) = 0.5691$, $4\pi\varepsilon_0\Phi(0.5, 0.25) = 0.7205$, $4\pi\varepsilon_0\Phi(0.5, 0.5) = 0.9258$. Comment on rate of convergence and final accuracy.

2.2 Solution

Let

$$u(x, y) \equiv 4\pi\varepsilon_0\, \Phi(x, y).$$

Then Poisson's equation becomes

$$\nabla^2 u = -4\pi,$$

since the charge density is constant and equal to 1 in the interior.

With $h = 0.25$, symmetry reduces the nine interior points to the three unknowns

$$a = u(0.25, 0.25), \qquad b = u(0.5, 0.25), \qquad c = u(0.5, 0.5).$$

All boundary values are zero.

Using the improved-grid 9-point formula,

$$\frac{1}{6h^2}\Big[4\big(u_E + u_W + u_N + u_S\big) + \big(u_{NE} + u_{NW} + u_{SE} + u_{SW}\big) - 20u_0\Big] \approx \nabla^2 u,$$

and substituting $\nabla^2 u = -4\pi$, we solve for the central value:

$$u_0 = \frac{1}{5}(u_E + u_W + u_N + u_S) + \frac{1}{20}(u_{NE} + u_{NW} + u_{SE} + u_{SW}) + \frac{6\pi h^2}{5}.$$

For $h = 0.25$,

$$d \equiv \frac{6\pi h^2}{5} = \frac{3\pi}{40} \approx 0.235619449.$$

By symmetry, the three update equations are

$$a = 0.4b + 0.05c + d,$$
$$b = \frac{0.4a + 0.2c + d}{0.9},$$
$$c = 0.2a + 0.8b + d.$$

We start with

$$a^{(0)} = b^{(0)} = c^{(0)} = 1.0.$$

For Jacobi iteration,

$$a^{(n+1)} = 0.4b^{(n)} + 0.05c^{(n)} + d,$$
$$b^{(n+1)} = \frac{0.4a^{(n)} + 0.2c^{(n)} + d}{0.9},$$
$$c^{(n+1)} = 0.2a^{(n)} + 0.8b^{(n)} + d.$$

The first ten Jacobi iterations are

n	$a^{(n)}$	$b^{(n)}$	$c^{(n)}$
1	0.685619	0.928466	1.235619
2	0.668787	0.841101	1.115516
3	0.627836	0.806930	1.042258
4	0.610505	0.772450	1.006731
5	0.594936	0.756853	0.975681
6	0.587145	0.743033	0.960089
7	0.580837	0.736106	0.947475
8	0.577435	0.730499	0.940671
9	0.574853	0.727475	0.935506
10	0.573385	0.725180	0.932570

For Gauss–Seidel iteration, using the order $a \to b \to c$ in each sweep,

$$a^{(n+1)} = 0.4b^{(n)} + 0.05c^{(n)} + d,$$
$$b^{(n+1)} = \frac{0.4a^{(n+1)} + 0.2c^{(n)} + d}{0.9},$$
$$c^{(n+1)} = 0.2a^{(n+1)} + 0.8b^{(n+1)} + d.$$

The first ten Gauss–Seidel iterations are

n	$a^{(n)}$	$b^{(n)}$	$c^{(n)}$
1	0.685619	0.788741	0.998736
2	0.601553	0.742666	0.949061
3	0.580639	0.730882	0.936453
4	0.575245	0.727807	0.933913
5	0.574337	0.727228	0.933267
6	0.574183	0.727120	0.933138
7	0.574157	0.727099	0.933112
8	0.574152	0.727095	0.933107
9	0.574151	0.727095	0.933106
10	0.574151	0.727094	0.933106

The exact values given in the problem are

$$a_{\text{exact}} = 0.5691, \qquad b_{\text{exact}} = 0.7205, \qquad c_{\text{exact}} = 0.9258.$$

Thus at the tenth iteration,

$$\text{Jacobi: } (a, b, c) = (0.573385,\ 0.725180,\ 0.932570),$$
$$\text{Gauss–Seidel: } (a, b, c) = (0.574151,\ 0.727094,\ 0.933106).$$

The absolute errors relative to the exact continuum values are

$$\text{Jacobi: } (0.004285,\ 0.004680,\ 0.006770),$$
$$\text{Gauss–Seidel: } (0.005051,\ 0.006594,\ 0.007306).$$

Therefore Gauss–Seidel converges much faster to the discrete improved-grid solution, reaching near-stationarity after about five or six sweeps, whereas Jacobi converges more slowly. The limiting discrete solution is not exactly equal to the quoted exact continuum solution because the finite mesh still introduces discretization error.

Problem 1.2

3.1 Problem Statement

The Dirac delta function in three dimensions can be taken as the improper limit as $\alpha \to 0$ of the Gaussian function

$$D(\alpha; x, y, z) = (2\pi)^{-3/2}\alpha^{-3}\exp\left[-\frac{1}{2\alpha^2}(x^2+y^2+z^2)\right].$$

Consider a general orthogonal coordinate system specified by the surfaces $u =$ constant, $v =$ constant, $w =$ constant, with length elements du/U, dv/V, dw/W in the three perpendicular directions. Show that

$$\delta(\mathbf{x}-\mathbf{x}') = \delta(u-u')\,\delta(v-v')\,\delta(w-w')\,UVW$$

by considering the limit of the Gaussian above. Note that as $\alpha \to 0$ only the infinitesimal length element need be used for the distance between the points in the exponent.

3.2 Solution

We begin from the standard Gaussian representation of the three-dimensional delta function:

$$\delta(\mathbf{x}-\mathbf{x}') = \lim_{\alpha\to 0}(2\pi)^{-3/2}\alpha^{-3}\exp\left[-\frac{|\mathbf{x}-\mathbf{x}'|^2}{2\alpha^2}\right].$$

We now express this in a general orthogonal curvilinear coordinate system (u, v, w).

In such a system the infinitesimal line element is

$$ds^2 = \left(\frac{du}{U}\right)^2 + \left(\frac{dv}{V}\right)^2 + \left(\frac{dw}{W}\right)^2.$$

Hence, for two nearby points (u, v, w) and (u', v', w'), we may write

$$|\mathbf{x}-\mathbf{x}'|^2 \approx \frac{(u-u')^2}{U^2} + \frac{(v-v')^2}{V^2} + \frac{(w-w')^2}{W^2},$$

where U, V, W are evaluated at the point $\mathbf{x}'$ (or equivalently at $\mathbf{x}$ in the limit $\alpha \to 0$). This is exactly the approximation the problem instructs us to use: as $\alpha \to 0$, only the infinitesimal distance contributes in the exponent.

Therefore,

$$\delta(\mathbf{x}-\mathbf{x}') = \lim_{\alpha\to 0}(2\pi)^{-3/2}\alpha^{-3}\exp\left[-\frac{1}{2\alpha^2}\left(\frac{(u-u')^2}{U^2} + \frac{(v-v')^2}{V^2} + \frac{(w-w')^2}{W^2}\right)\right].$$

Since the exponent is a sum of three independent quadratic terms, the Gaussian factorizes:

$$\delta(\mathbf{x}-\mathbf{x}') = \lim_{\alpha\to 0}\prod_{\xi=u,v,w}\left[(2\pi)^{-1/2}\alpha^{-1}\exp\left(-\frac{(\xi-\xi')^2}{2\alpha^2 H_\xi^2}\right)\right],$$

where

$$H_u = \frac{1}{U}, \qquad H_v = \frac{1}{V}, \qquad H_w = \frac{1}{W}.$$

Writing this out explicitly,

$$\delta(\mathbf{x}-\mathbf{x}') = \lim_{\alpha\to 0}\left[(2\pi)^{-1/2}\alpha^{-1}e^{-\frac{(u-u')^2}{2\alpha^2U^2}}\right]\left[(2\pi)^{-1/2}\alpha^{-1}e^{-\frac{(v-v')^2}{2\alpha^2V^2}}\right]\left[(2\pi)^{-1/2}\alpha^{-1}e^{-\frac{(w-w')^2}{2\alpha^2W^2}}\right].$$

Now use the one-dimensional scaling property of the delta function:

$$\delta(a\xi) = \frac{1}{|a|}\delta(\xi),$$

or equivalently, from the Gaussian representation,

$$\lim_{\alpha\to 0}(2\pi)^{-1/2}\alpha^{-1}\exp\left[-\frac{(\xi-\xi')^2}{2\alpha^2A^2}\right] = A\,\delta(\xi-\xi') \qquad (A>0).$$

Applying this to each factor gives

$$\lim_{\alpha\to 0}(2\pi)^{-1/2}\alpha^{-1}e^{-\frac{(u-u')^2}{2\alpha^2U^2}} = U\,\delta(u-u'),$$

$$\lim_{\alpha\to 0}(2\pi)^{-1/2}\alpha^{-1}e^{-\frac{(v-v')^2}{2\alpha^2V^2}} = V\,\delta(v-v'),$$

$$\lim_{\alpha\to 0}(2\pi)^{-1/2}\alpha^{-1}e^{-\frac{(w-w')^2}{2\alpha^2W^2}} = W\,\delta(w-w').$$

Multiplying the three limits together, we obtain

$$\delta(\mathbf{x}-\mathbf{x}') = U\,V\,W\,\delta(u-u')\,\delta(v-v')\,\delta(w-w').$$

Hence,

$$\boxed{\delta(\mathbf{x}-\mathbf{x}') = \delta(u-u')\,\delta(v-v')\,\delta(w-w')\,UVW}.$$

This is the desired result.

As a check, note that the volume element in these coordinates is

$$d\tau = \frac{du\,dv\,dw}{UVW},$$

so

$$\int \delta(\mathbf{x}-\mathbf{x}')\,d\tau = \int \delta(u-u')\delta(v-v')\delta(w-w')\,UVW\frac{du\,dv\,dw}{UVW} = 1,$$

which confirms that the normalization is correct.

Problem 1.23

4.1 Problem Statement

A transmission line consists of a long straight conductor with a hollow square region in its interior, with a square conductor of one-quarter the area of the hollow region centered in the empty space, with edges parallel to the inner sides of outer con- ductor. If the conductors are raised to different potentials, the potential and elec- tric field in the space between them exhibit an eightfold symmetry; the basic unit is sketched in the accompanying figure. The efficacy of the relaxation method in de- termining the properties of the transmission line can be illustrated by a simple calculation.

(a) Using only the four interior points indicated in the figure, write down the relaxation equation for each point for the "cross" and the "improved" averaging schemes (defined in Problem 1.22) if the inner conductor has $\Phi = 100$ V and the outer has $\Phi_0 = 0$. By performing either the relaxation iteration process or solving the set of algebraic equations for each scheme, find estimates for the potential at each of the four points for the two schemes.

(b) From the results of part **a** make the best estimate (or estimates) you can for the capacitance per unit length of the transmission line.

(c) (Optional) Using your favorite computational tools, repeat the relaxation cal- culation with half the lattice spacing (21 interior points) and compare.

4.2 Solution

Let the four unknown lattice potentials be $\Phi_1, \Phi_2, \Phi_3, \Phi_4$, and let the inner conductor be at

$$\Phi = 100 \text{ V},$$

while the outer conductor is at

$$\Phi_0 = 0.$$

Because of the eightfold symmetry, some neighbors of the interior points are mirror images of other interior points. That is why the same potential appears more than once in the averaging formulas.

Part (a): Cross averaging

For Laplace's equation, the cross averaging scheme is

$$\Phi_P = \frac{1}{4}\left(\Phi_N + \Phi_S + \Phi_E + \Phi_W\right).$$

Using the geometry of the four labeled points in the figure:

$$\Phi_1 = \frac{1}{4}\left(\Phi_2 + \Phi_2 + \Phi + \Phi_0\right) = \frac{1}{4}\left(2\Phi_2 + 100\right),$$

since one neighbor is the inner conductor, one is the outer conductor, and the two remaining neighbors are symmetry-equivalent to point 2.

$$\Phi_2 = \frac{1}{4}\left(\Phi_1 + \Phi_3 + \Phi + \Phi_0\right) = \frac{1}{4}\left(\Phi_1 + \Phi_3 + 100\right).$$

$$\Phi_3 = \frac{1}{4}\left(\Phi_2 + \Phi_4 + \Phi + \Phi_0\right) = \frac{1}{4}\left(\Phi_2 + \Phi_4 + 100\right).$$

$$\Phi_4 = \frac{1}{4}\left(\Phi_3 + \Phi_3 + \Phi_0 + \Phi_0\right) = \frac{1}{2}\Phi_3.$$

Thus the linear system is

$$\begin{aligned} 4\Phi_1 - 2\Phi_2 &= 100, \\ -\Phi_1 + 4\Phi_2 - \Phi_3 &= 100, \\ -\Phi_2 + 4\Phi_3 - \Phi_4 &= 100, \\ -\Phi_3 + 2\Phi_4 &= 0. \end{aligned}$$

From the last equation,

$$\Phi_4 = \frac{\Phi_3}{2}.$$

Substitute into the third equation:

$$-\Phi_2 + 4\Phi_3 - \frac{\Phi_3}{2} = 100 \quad \Rightarrow \quad -\Phi_2 + \frac{7}{2}\Phi_3 = 100.$$

From the first equation,

$$\Phi_1 = \frac{100 + 2\Phi_2}{4} = 25 + \frac{\Phi_2}{2}.$$

Insert this into the second equation:

$$-\left(25 + \frac{\Phi_2}{2}\right) + 4\Phi_2 - \Phi_3 = 100,$$

$$\frac{7}{2}\Phi_2 - \Phi_3 = 125.$$

So we now solve

$$\begin{aligned} \frac{7}{2}\Phi_2 - \Phi_3 &= 125, \\ -\Phi_2 + \frac{7}{2}\Phi_3 &= 100. \end{aligned}$$

Multiply the second equation by 7 and eliminate Φ_2:

$$-7\Phi_2 + \frac{49}{2}\Phi_3 = 700,$$

while the first gives

$$7\Phi_2 - 2\Phi_3 = 250.$$

Adding,

$$\frac{45}{2}\Phi_3 = 950 \quad \Rightarrow \quad \Phi_3 = \frac{1900}{45} = \frac{380}{9} \cdot \frac{1}{10} = 42.22 \text{ V}.$$

More exactly,

$$\Phi_3 = \frac{38}{90}(100) = 42.22 \text{ V}.$$

Then

$$\Phi_4 = \frac{\Phi_3}{2} = \frac{19}{90}(100) = 21.11 \text{ V}.$$

Next,

$$\frac{7}{2}\Phi_2 - \Phi_3 = 125 \quad \Rightarrow \quad \frac{7}{2}\Phi_2 = 125 + 42.22 = 167.22,$$

so

$$\Phi_2 = \frac{43}{90}(100) = 47.78 \text{ V}.$$

Finally,

$$\Phi_1 = 25 + \frac{\Phi_2}{2} = 25 + \frac{47.78}{2} = 48.89 \text{ V},$$

so

$$\Phi_1 = \frac{44}{90}(100) = 48.89 \text{ V}.$$

Therefore, for the cross scheme,

$$\boxed{\Phi_1 = 48.89 \text{ V}, \quad \Phi_2 = 47.78 \text{ V}, \quad \Phi_3 = 42.22 \text{ V}, \quad \Phi_4 = 21.11 \text{ V}.}$$

Part (a): Improved averaging

For the improved scheme of Problem 1.22, the potential is taken as

$$\Phi_P = \frac{1}{5}\left(S_c + \frac{1}{4}S_s\right),$$

where S_c is the cross sum of the four nearest neighbors and S_s is the sum over the four diagonal neighbors.

Applying this to the four labeled points and using symmetry gives the linear system

$$5\Phi_1 - 2\Phi_2 = 150,$$

$$-\Phi_1 + 5\Phi_2 - \Phi_3 = 150,$$

$$-\Phi_2 + \frac{19}{4}\Phi_3 - \Phi_4 = 125,$$

$$-2\Phi_3 + 5\Phi_4 = 25.$$

We solve this step by step.

From the first equation,

$$\Phi_1 = 30 + \frac{2}{5}\Phi_2.$$

From the fourth equation,

$$\Phi_4 = 5 + \frac{2}{5}\Phi_3.$$

Insert these into the second and third equations.

The second becomes

$$-\left(30 + \frac{2}{5}\Phi_2\right) + 5\Phi_2 - \Phi_3 = 150,$$

$$\frac{23}{5}\Phi_2 - \Phi_3 = 180.$$

The third becomes

$$-\Phi_2 + \frac{19}{4}\Phi_3 - \left(5 + \frac{2}{5}\Phi_3\right) = 125,$$

$$-\Phi_2 + \left(\frac{19}{4} - \frac{2}{5}\right)\Phi_3 = 130.$$

Now,

$$\frac{19}{4} - \frac{2}{5} = \frac{95 - 8}{20} = \frac{87}{20},$$

so

$$-\Phi_2 + \frac{87}{20}\Phi_3 = 130.$$

Thus we solve

$$\frac{23}{5}\Phi_2 - \Phi_3 = 180,$$

$$-\Phi_2 + \frac{87}{20}\Phi_3 = 130.$$

From the first,

$$\Phi_2 = \frac{5}{23}(180 + \Phi_3).$$

Substitute into the second:

$$-\frac{5}{23}(180 + \Phi_3) + \frac{87}{20}\Phi_3 = 130.$$

Multiply through by $460 = 20 \times 23$:

$$-100(180 + \Phi_3) + 2001\Phi_3 = 59800.$$

Hence

$$-18000 - 100\Phi_3 + 2001\Phi_3 = 59800,$$

$$1901\Phi_3 = 77800,$$

$$\Phi_3 \approx 40.93 \text{ V}.$$

Then

$$\Phi_4 = 5 + \frac{2}{5}\Phi_3 \approx 5 + \frac{2}{5}(40.93) \approx 21.37 \text{ V}.$$

Also,

$$\Phi_2 = \frac{5}{23}(180 + 40.93) \approx 48.03 \text{ V},$$

and

$$\Phi_1 = 30 + \frac{2}{5}\Phi_2 \approx 30 + \frac{2}{5}(48.03) \approx 49.21 \text{ V}.$$

Therefore, for the improved scheme,

$$\boxed{\Phi_1 = 49.21 \text{ V}, \quad \Phi_2 = 48.03 \text{ V}, \quad \Phi_3 = 40.93 \text{ V}, \quad \Phi_4 = 21.37 \text{ V}.}$$

Part (b): Capacitance per unit length

The capacitance per unit length is

$$C = \frac{Q}{\Phi},$$

where Q is the total charge per unit length on the inner conductor and

$$\Phi = 100 \text{ V}$$

is the potential difference.

Because of the fourfold symmetry of the conductor, it is enough to compute the charge on one side of the inner conductor and then multiply by 4.

The surface charge density is related to the normal electric field by

$$\sigma = \varepsilon_0 E_n = \varepsilon_0 \frac{\partial \Phi}{\partial n}.$$

On the discrete lattice, the normal derivative at the conductor surface is approximated by

$$\frac{\partial \Phi}{\partial n} \approx \frac{\Phi - \Phi_{\text{adjacent}}}{h}.$$

When multiplied by the segment length h, the contribution of one lattice segment to the charge per unit length is simply

$$\Delta Q = \varepsilon_0(\Phi - \Phi_{\text{adjacent}}).$$

Along one side of the inner conductor, there are contributions corresponding to the adjacent points $\Phi_1, \Phi_2, \Phi_2, \Phi_3, \Phi_3$, so

$$Q_{\text{one side}} = \varepsilon_0 \left(5\Phi - \Phi_1 - 2\Phi_2 - 2\Phi_3\right).$$

Hence the total charge per unit length is

$$Q_{\text{tot}} = 4Q_{\text{one side}},$$

and therefore

$$C = \frac{Q_{\text{tot}}}{\Phi} = \frac{4}{\Phi}\,\varepsilon_0 \left(5\Phi - \Phi_1 - 2\Phi_2 - 2\Phi_3\right).$$

Cross scheme:

$$Q_c = \varepsilon_0 \left[5(100) - 48.89 - 2(47.78) - 2(42.22)\right] = 271.11\,\varepsilon_0.$$

Thus

$$C_c = \frac{4Q_c}{100} = \frac{4(271.11\,\varepsilon_0)}{100} \approx 10.84\,\varepsilon_0 \text{ F/m}.$$

Improved scheme:

$$Q_i = \varepsilon_0 \left[5(100) - 49.21 - 2(48.03) - 2(40.93)\right] = 272.87\,\varepsilon_0.$$

Thus

$$C_i = \frac{4Q_i}{100} = \frac{4(272.87\,\varepsilon_0)}{100} \approx 10.91\,\varepsilon_0 \text{ F/m}.$$

Therefore the coarse-grid estimates for the capacitance per unit length are

$$\boxed{C_c \approx 10.84\,\varepsilon_0 \text{ F/m}}, \qquad \boxed{C_i \approx 10.91\,\varepsilon_0 \text{ F/m}}.$$

A reasonable best estimate from these two coarse-grid results is therefore

$$\boxed{C \approx 10.9\,\varepsilon_0 \text{ F/m}.}$$

The improved averaging scheme gives slightly different node values and a slightly larger capacitance estimate, but both schemes are already fairly close to the more accurate value obtained from a much finer grid.

Problem 1.3

5.1 Problem Statement

Using Dirac delta functions in the appropriate coordinates, express the following charge distributions as three-dimensional charge densities $\rho(\mathbf{x})$.

(a) In spherical coordinates, a charge Q uniformly distributed over a spherical shell of radius R.

(b) In cylindrical coordinates, a charge λ per unit length uniformly distributed over a cylindrical surface of radius b.

(c) In cylindrical coordinates, a charge Q spread uniformly over a flat circular disc of negligible thickness and radius R.

(d) The same as part **(c)**, but using spherical coordinates.

5.2 Solution

The general idea is that a singular charge distribution concentrated on a surface or line can be represented in three dimensions by inserting Dirac delta functions that localize the charge in the appropriate coordinate, and then choosing the coefficient so that the integral of ρ over all space gives the correct total charge.

We will use the volume elements

$$d\tau = r^2 \sin\theta\, dr\, d\theta\, d\phi \qquad \text{(spherical)},$$

and

$$d\tau = s\, ds\, d\phi\, dz \qquad \text{(cylindrical)},$$

where I use s for the cylindrical radial coordinate.

We also use the basic property of the delta function:

$$\int_{-\infty}^{\infty} \delta(x-a) f(x)\, dx = f(a).$$

(a) Spherical shell of radius R carrying total charge Q

The charge is uniformly distributed over the spherical surface

$$r = R.$$

So the density must vanish everywhere except at $r = R$, which means it must contain a factor $\delta(r - R)$.

Let

$$\rho(\mathbf{x}) = A\,\delta(r - R),$$

where A is to be determined.

Since the total charge must be Q,

$$Q = \iiint \rho(\mathbf{x})\,d\tau.$$

Substitute the trial form:

$$Q = \iiint A\,\delta(r - R)\,r^2 \sin\theta\,dr\,d\theta\,d\phi.$$

Separate the integrals:

$$Q = A\left(\int_0^\infty \delta(r - R)r^2\,dr\right)\left(\int_0^\pi \sin\theta\,d\theta\right)\left(\int_0^{2\pi} d\phi\right).$$

Now,

$$\int_0^\infty \delta(r - R)r^2\,dr = R^2,$$

$$\int_0^\pi \sin\theta\,d\theta = 2,$$

$$\int_0^{2\pi} d\phi = 2\pi.$$

Therefore

$$Q = A(R^2)(2)(2\pi) = 4\pi A R^2.$$

Hence

$$A = \frac{Q}{4\pi R^2}.$$

So the required charge density is

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{4\pi R^2}\,\delta(r - R)}.$$

This is exactly what we expect physically: the factor $Q/(4\pi R^2)$ is the uniform surface charge density on the sphere, and the delta function localizes it to the surface.

(b) Cylindrical surface of radius b carrying charge λ per unit length

Now the charge is uniformly distributed on the cylindrical surface

$$s = b,$$

with charge per unit length equal to λ.

Because the distribution extends indefinitely along z, the total charge is infinite, so the correct normalization condition is that the charge in a length L of the cylinder must be

$$Q_L = \lambda L.$$

Let

$$\rho(\mathbf{x}) = A\,\delta(s - b).$$

Then the charge contained in a segment of length L is

$$Q_L = \int_0^L \int_0^{2\pi} \int_0^\infty A\,\delta(s - b)\,s\,ds\,d\phi\,dz.$$

Separate the integrals:

$$Q_L = A\left(\int_0^\infty \delta(s - b)s\,ds\right)\left(\int_0^{2\pi} d\phi\right)\left(\int_0^L dz\right).$$

These give

$$\int_0^\infty \delta(s-b)s\,ds = b, \qquad \int_0^{2\pi} d\phi = 2\pi, \qquad \int_0^L dz = L.$$

Hence

$$Q_L = A(b)(2\pi)(L) = 2\pi bAL.$$

But we require

$$Q_L = \lambda L.$$

Therefore

$$2\pi bAL = \lambda L \quad \Rightarrow \quad A = \frac{\lambda}{2\pi b}.$$

Thus

$$\boxed{\rho(\mathbf{x}) = \frac{\lambda}{2\pi b}\,\delta(s-b)}.$$

The coefficient $\lambda/(2\pi b)$ is precisely the uniform surface charge density on the cylindrical surface.

(c) Flat circular disc of radius R in cylindrical coordinates

The disc lies in the plane

$$z = 0,$$

and occupies the region

$$0 \le s \le R.$$

The charge is spread uniformly over the disc, so the surface charge density is

$$\sigma = \frac{Q}{\pi R^2}.$$

Since the disc has negligible thickness, the three-dimensional density must be concentrated at $z = 0$, so it must contain a factor $\delta(z)$. Also, the charge should exist only for $s \le R$, so we multiply by a step-function restriction. Using the Heaviside function $\Theta(R-s)$, we write

$$\rho(\mathbf{x}) = \sigma\,\delta(z)\,\Theta(R-s).$$

Substituting σ,

$$\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\delta(z)\,\Theta(R-s).$$

Let us verify the normalization:

$$\iiint \rho(\mathbf{x})\,d\tau = \int_{-\infty}^{\infty}\int_0^{2\pi}\int_0^{\infty} \frac{Q}{\pi R^2}\,\delta(z)\,\Theta(R-s)\,s\,ds\,d\phi\,dz.$$

Because of $\Theta(R-s)$, the radial integral runs only from 0 to R:

$$= \frac{Q}{\pi R^2}\left(\int_{-\infty}^{\infty}\delta(z)\,dz\right)\left(\int_0^{2\pi} d\phi\right)\left(\int_0^R s\,ds\right).$$

Now

$$\int_{-\infty}^{\infty}\delta(z)\,dz = 1, \qquad \int_0^{2\pi} d\phi = 2\pi, \qquad \int_0^R s\,ds = \frac{R^2}{2}.$$

Therefore

$$\iiint \rho(\mathbf{x})\,d\tau = \frac{Q}{\pi R^2}(1)(2\pi)\left(\frac{R^2}{2}\right) = Q.$$

So the normalization is correct.

Hence the charge density is

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\delta(z)\,\Theta(R-s)}.$$

(d) The same disc, but written in spherical coordinates

Now we describe the same flat disc using spherical coordinates. The disc still lies in the plane

$$z = 0.$$

In spherical coordinates,

$$z = r\cos\theta.$$

So localization to the plane requires

$$\delta(z) = \delta(r\cos\theta).$$

Also, the condition that the point lie inside the radius-R disc is

$$s \le R.$$

But in spherical coordinates,

$$s = r\sin\theta.$$

Hence the restriction becomes

$$r\sin\theta \le R,$$

which can be written with a step function as

$$\Theta(R - r\sin\theta).$$

Therefore a natural spherical-coordinate representation is

$$\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\delta(r\cos\theta)\,\Theta(R - r\sin\theta).$$

So one acceptable answer is

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\delta(r\cos\theta)\,\Theta(R - r\sin\theta)}.$$

We can also rewrite this in a more explicit form using the identity

$$\delta(f(\theta)) = \sum_i \frac{\delta(\theta - \theta_i)}{|f'(\theta_i)|}.$$

For

$$f(\theta) = r\cos\theta,$$

the zero in the interval $0 \le \theta \le \pi$ is at

$$\theta = \frac{\pi}{2},$$

and

$$\frac{d}{d\theta}(r\cos\theta)\bigg|_{\theta=\pi/2} = -r\sin\frac{\pi}{2} = -r.$$

Hence

$$\delta(r\cos\theta) = \frac{1}{r}\,\delta\Big(\theta - \frac{\pi}{2}\Big).$$

Also, when $\theta = \pi/2$,

$$r\sin\theta = r,$$

so the radial restriction becomes simply

$$\Theta(R - r).$$

Thus an equivalent form is

$$\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\frac{1}{r}\,\delta\Big(\theta - \frac{\pi}{2}\Big)\,\Theta(R - r).$$

Therefore an equivalent spherical-coordinate expression is

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{\pi R^2 r}\,\delta\Big(\theta - \frac{\pi}{2}\Big)\,\Theta(R - r)}.$$

Let us check this normalization:

$$\iiint \rho(\mathbf{x})\,d\tau = \int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} \frac{Q}{\pi R^2 r}\,\delta\Big(\theta - \frac{\pi}{2}\Big)\,\Theta(R - r)\,r^2\sin\theta\,dr\,d\theta\,d\phi.$$

This becomes

$$\frac{Q}{\pi R^2}\left(\int_0^{2\pi} d\phi\right)\left(\int_0^{\pi} \delta\Big(\theta - \frac{\pi}{2}\Big)\sin\theta\, d\theta\right)\left(\int_0^{R} r\, dr\right).$$

Now

$$\int_0^{2\pi} d\phi = 2\pi, \qquad \int_0^{\pi} \delta\Big(\theta - \frac{\pi}{2}\Big)\sin\theta\, d\theta = 1, \qquad \int_0^{R} r\, dr = \frac{R^2}{2}.$$

Hence

$$\iiint \rho(\mathbf{x})\, d\tau = \frac{Q}{\pi R^2}(2\pi)(1)\left(\frac{R^2}{2}\right) = Q,$$

as required.

So the final answers are

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{4\pi R^2}\,\delta(r - R)}$$

for part **(a)**,

$$\boxed{\rho(\mathbf{x}) = \frac{\lambda}{2\pi b}\,\delta(s - b)}$$

for part **(b)**,

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\delta(z)\,\Theta(R - s)}$$

for part **(c)**, and

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{\pi R^2 r}\,\delta\Big(\theta - \frac{\pi}{2}\Big)\,\Theta(R - r)}$$

for part **(d)**, equivalently

$$\boxed{\rho(\mathbf{x}) = \frac{Q}{\pi R^2}\,\delta(r\cos\theta)\,\Theta(R - r\sin\theta)}.$$

Problem 1.22

6.1 Problem Statement

Two-dimensional relaxation calculations commonly use sites on a square lattice with spacing $\Delta x = \Delta y = h$, and label the sites by (i, j), where i, j are integers and $x_i = ih + x_0$, $y_j = jh + y_0$. The value of the potential at (i, j) can be approximated by the average of the values at neighboring sites. [Recall the relevant theorem about harmonic functions.] But what average?

(a) If $F(x, y)$ is a well-behaved function in the neighborhood of the origin, but not necessarily harmonic, by explicit Taylor series expansions, show that the "cross" sum

$$S_c = F(h, 0) + F(0, h) + F(-h, 0) + F(0, -h)$$

can be expressed as

$$S_c = 4F(0, 0) + h^2\nabla^2 F + \frac{h^4}{12}\left(F_{xxxx} + F_{yyyy}\right) + O(h^6).$$

(b) Similarly, show that the "square" sum,

$$S_s = F(h, h) + F(-h, h) + F(-h, -h) + F(h, -h)$$

can be expressed as

$$S_s = 4F(0, 0) + 2h^2\nabla^2 F - \frac{h^4}{3}\left(F_{xxxx} + F_{yyyy}\right) + h^4 F_{xxyy} + O(h^6),$$

or equivalently,

$$S_s = 4F(0, 0) + 2h^2\nabla^2 F - \frac{h^4}{3}\left(F_{xxxx} + F_{yyyy}\right) + \frac{h^4}{2}\nabla^2(\nabla^2 F) + O(h^6).$$

Here F_{xxxx} is the fourth partial derivative of F with respect to x, evaluated at $x = 0$, $y = 0$, etc. If $\nabla^2 F = 0$, the averages $S_c/4$ and $S_s/4$ each give the value of $F(0,0)$, correct to order h^3 inclusive. Note that an improvement can be obtained by forming the "improved" average,

$$\langle\langle F(0,0)\rangle\rangle = \frac{1}{5}\left[S_c + \frac{1}{4}S_s\right],$$

where

$$\langle\langle F(0,0)\rangle\rangle = F(0,0) + \frac{3}{10}h^2\nabla^2 F + \frac{h^4}{40}\nabla^2(\nabla^2 F) + O(h^6).$$

If $\nabla^2 F = 0$, then $\langle\langle F(0,0)\rangle\rangle$ gives $F(0,0)$, correct to order h^5 inclusive. For Poisson's equation, the charge density and its lowest order Laplacian can be inserted for the same accuracy.

6.2 Solution

We expand $F(x,y)$ in a Taylor series about the origin. Since all evaluations are made at points whose coordinates are $\pm h$ or 0, it is convenient to write the full two-variable Taylor expansion up to fourth order:

$$F(x,y) = F_0 + xF_x + yF_y + \frac{1}{2}\left(x^2F_{xx} + 2xyF_{xy} + y^2F_{yy}\right)$$
$$+\frac{1}{6}\left(x^3F_{xxx} + 3x^2yF_{xxy} + 3xy^2F_{xyy} + y^3F_{yyy}\right)$$
$$+\frac{1}{24}\left(x^4F_{xxxx} + 4x^3yF_{xxxy} + 6x^2y^2F_{xxyy} + 4xy^3F_{xyyy} + y^4F_{yyyy}\right) + O(h^5),$$

where all derivatives are evaluated at $(0,0)$, and we write

$$F_0 \equiv F(0,0).$$

Because the sums in the problem involve symmetrically placed points, all odd powers will cancel. That is the key reason these averaging schemes work well.

(a) The cross sum S_c

We first expand the four terms:

$$F(h,0) = F_0 + hF_x + \frac{h^2}{2}F_{xx} + \frac{h^3}{6}F_{xxx} + \frac{h^4}{24}F_{xxxx} + O(h^5),$$
$$F(-h,0) = F_0 - hF_x + \frac{h^2}{2}F_{xx} - \frac{h^3}{6}F_{xxx} + \frac{h^4}{24}F_{xxxx} + O(h^5),$$
$$F(0,h) = F_0 + hF_y + \frac{h^2}{2}F_{yy} + \frac{h^3}{6}F_{yyy} + \frac{h^4}{24}F_{yyyy} + O(h^5),$$
$$F(0,-h) = F_0 - hF_y + \frac{h^2}{2}F_{yy} - \frac{h^3}{6}F_{yyy} + \frac{h^4}{24}F_{yyyy} + O(h^5).$$

Now add them:

$$S_c = F(h,0) + F(-h,0) + F(0,h) + F(0,-h).$$

The linear terms cancel:

$$hF_x - hF_x = 0, \qquad hF_y - hF_y = 0.$$

The cubic terms also cancel:

$$\frac{h^3}{6}F_{xxx} - \frac{h^3}{6}F_{xxx} = 0, \qquad \frac{h^3}{6}F_{yyy} - \frac{h^3}{6}F_{yyy} = 0.$$

So we are left with

$$S_c = 4F_0 + h^2(F_{xx} + F_{yy}) + \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + O(h^6).$$

Since

$$\nabla^2 F = F_{xx} + F_{yy},$$

we obtain

$$\boxed{S_c = 4F(0,0) + h^2\nabla^2 F + \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + O(h^6)}.$$

That proves part **(a)**.

(b) The square sum S_s

Now consider the four corner points:

$$S_s = F(h,h) + F(-h,h) + F(-h,-h) + F(h,-h).$$

We expand each term to fourth order.
For (h,h):

$$F(h,h) = F_0 + hF_x + hF_y + \frac{h^2}{2}(F_{xx} + 2F_{xy} + F_{yy})$$
$$+\frac{h^3}{6}(F_{xxx} + 3F_{xxy} + 3F_{xyy} + F_{yyy})$$
$$+\frac{h^4}{24}(F_{xxxx} + 4F_{xxxy} + 6F_{xxyy} + 4F_{xyyy} + F_{yyyy}) + O(h^5).$$

For $(-h,h)$:

$$F(-h,h) = F_0 - hF_x + hF_y + \frac{h^2}{2}(F_{xx} - 2F_{xy} + F_{yy})$$
$$+\frac{h^3}{6}(-F_{xxx} + 3F_{xxy} - 3F_{xyy} + F_{yyy})$$
$$+\frac{h^4}{24}(F_{xxxx} - 4F_{xxxy} + 6F_{xxyy} - 4F_{xyyy} + F_{yyyy}) + O(h^5).$$

For $(-h,-h)$:

$$F(-h,-h) = F_0 - hF_x - hF_y + \frac{h^2}{2}(F_{xx} + 2F_{xy} + F_{yy})$$
$$+\frac{h^3}{6}(-F_{xxx} - 3F_{xxy} - 3F_{xyy} - F_{yyy})$$
$$+\frac{h^4}{24}(F_{xxxx} + 4F_{xxxy} + 6F_{xxyy} + 4F_{xyyy} + F_{yyyy}) + O(h^5).$$

For $(h,-h)$:

$$F(h,-h) = F_0 + hF_x - hF_y + \frac{h^2}{2}(F_{xx} - 2F_{xy} + F_{yy})$$
$$+\frac{h^3}{6}(F_{xxx} - 3F_{xxy} + 3F_{xyy} - F_{yyy})$$
$$+\frac{h^4}{24}(F_{xxxx} - 4F_{xxxy} + 6F_{xxyy} - 4F_{xyyy} + F_{yyyy}) + O(h^5).$$

Now add all four.

Zeroth order:

$$4F_0.$$

First order: All linear terms cancel by symmetry.

Second order: The F_{xy} terms cancel, while the F_{xx} and F_{yy} terms add:

$$4 \cdot \frac{h^2}{2} F_{xx} = 2h^2 F_{xx}, \qquad 4 \cdot \frac{h^2}{2} F_{yy} = 2h^2 F_{yy}.$$

Thus the second-order contribution is

$$2h^2(F_{xx} + F_{yy}) = 2h^2 \nabla^2 F.$$

Third order: All cubic terms cancel by symmetry.

Fourth order: The mixed odd terms F_{xxxy} and F_{xyyy} cancel. The remaining terms add:

$$4 \cdot \frac{h^4}{24} F_{xxxx} = \frac{h^4}{6} F_{xxxx},$$

$$4 \cdot \frac{h^4}{24} F_{yyyy} = \frac{h^4}{6} F_{yyyy},$$

$$4 \cdot \frac{6h^4}{24} F_{xxyy} = h^4 F_{xxyy}.$$

Hence

$$S_s = 4F_0 + 2h^2(F_{xx} + F_{yy}) + \frac{h^4}{6}(F_{xxxx} + F_{yyyy}) + h^4 F_{xxyy} + O(h^6).$$

So one form is

$$\boxed{S_s = 4F(0,0) + 2h^2\nabla^2 F + \frac{h^4}{6}(F_{xxxx} + F_{yyyy}) + h^4 F_{xxyy} + O(h^6)}.$$

Now we rewrite it in the form stated in the problem. Observe that

$$\nabla^2(\nabla^2 F) = (F_{xx} + F_{yy})_{xx} + (F_{xx} + F_{yy})_{yy} = F_{xxxx} + 2F_{xxyy} + F_{yyyy}.$$

Therefore

$$F_{xxyy} = \frac{1}{2}\left[\nabla^2(\nabla^2 F) - F_{xxxx} - F_{yyyy}\right].$$

Substitute this into the expression for S_s:

$$S_s = 4F_0 + 2h^2\nabla^2 F + \frac{h^4}{6}(F_{xxxx} + F_{yyyy}) + \frac{h^4}{2}\left[\nabla^2(\nabla^2 F) - F_{xxxx} - F_{yyyy}\right] + O(h^6).$$

Now combine the coefficients of $F_{xxxx} + F_{yyyy}$:

$$\frac{1}{6} - \frac{1}{2} = -\frac{1}{3}.$$

Hence

$$\boxed{S_s = 4F(0,0) + 2h^2\nabla^2 F - \frac{h^4}{3}(F_{xxxx} + F_{yyyy}) + \frac{h^4}{2}\nabla^2(\nabla^2 F) + O(h^6)}.$$

This is exactly the required form.

Harmonic case

If F is harmonic, then

$$\nabla^2 F = 0.$$

So from part **(a)**,

$$S_c = 4F_0 + \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + O(h^6),$$

and therefore

$$\frac{S_c}{4} = F_0 + \frac{h^4}{48}(F_{xxxx} + F_{yyyy}) + O(h^6).$$

Similarly, from part **(b)**,

$$S_s = 4F_0 - \frac{h^4}{3}(F_{xxxx} + F_{yyyy}) + \frac{h^4}{2}\nabla^2(\nabla^2 F) + O(h^6).$$

But if $\nabla^2 F = 0$, then also

$$\nabla^2(\nabla^2 F) = 0.$$

Hence

$$S_s = 4F_0 - \frac{h^4}{3}(F_{xxxx} + F_{yyyy}) + O(h^6),$$

so

$$\frac{S_s}{4} = F_0 - \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + O(h^6).$$

Thus both $S_c/4$ and $S_s/4$ reproduce $F(0,0)$ with an error of order h^4, so they are correct through order h^3 inclusive.

Improved average

Now define

$$\langle\langle F(0,0)\rangle\rangle = \frac{1}{5}\left[S_c + \frac{1}{4}S_s\right].$$

Insert the expansions for S_c and S_s.

First,

$$S_c = 4F_0 + h^2\nabla^2 F + \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + O(h^6),$$

and

$$\frac{1}{4}S_s = F_0 + \frac{h^2}{2}\nabla^2 F - \frac{h^4}{12}(F_{xxxx} + F_{yyyy}) + \frac{h^4}{8}\nabla^2(\nabla^2 F) + O(h^6).$$

Add them:

$$S_c + \frac{1}{4}S_s = 5F_0 + \frac{3}{2}h^2\nabla^2 F + \frac{h^4}{8}\nabla^2(\nabla^2 F) + O(h^6),$$

because the fourth-derivative terms

$$\frac{h^4}{12}(F_{xxxx} + F_{yyyy})$$

and

$$-\frac{h^4}{12}(F_{xxxx} + F_{yyyy})$$

cancel exactly.

Now divide by 5:

$$\boxed{\langle\langle F(0,0)\rangle\rangle = F(0,0) + \frac{3}{10}h^2\nabla^2 F + \frac{h^4}{40}\nabla^2(\nabla^2 F) + O(h^6)}.$$

This is the desired improved formula.

If F is harmonic, $\nabla^2 F = 0$, and then

$$\nabla^2(\nabla^2 F) = 0$$

as well, so

$$\langle\langle F(0,0)\rangle\rangle = F(0,0) + O(h^6).$$

Therefore the improved average gives $F(0,0)$ correct through order h^5 inclusive.

Remark for Poisson's equation

If instead F satisfies Poisson's equation,

$$\nabla^2 F = \text{known source},$$

then the improved average remains useful because the $h^2\nabla^2 F$ term and, if needed, the $h^4\nabla^2(\nabla^2 F)$ term can be replaced directly by the source and its Laplacian. That is why this combination is especially effective in relaxation methods for electrostatics: it improves the local approximation while still using only nearby lattice values.

Problem 1.4

7.1 Problem Statement

Each of three charged spheres of radius a, one conducting, one having a uniform charge density within its volume, and one having a spherically symmetric charge density that varies radially as r^n ($n > -3$), has a total charge Q. Use Gauss's theorem to obtain the electric fields both inside and outside each sphere. Sketch the behavior of the fields as a function of radius for the first two spheres, and for the third with $n = -2, +2$.

7.2 Solution

Because all three charge distributions are spherically symmetric, the electric field must have the form

$$\mathbf{E}(r) = E(r)\,\hat{\mathbf{r}}.$$

That is, the field is purely radial and its magnitude depends only on the distance r from the center.

For any spherical Gaussian surface of radius r, Gauss's law gives

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}(r)}{\varepsilon_0}.$$

Since $\mathbf{E}$ is radial and has constant magnitude on a sphere of radius r,

$$\oint \mathbf{E} \cdot d\mathbf{a} = E(r) \oint da = E(r)\,4\pi r^2.$$

Therefore

$$E(r)\,4\pi r^2 = \frac{Q_{\text{enc}}(r)}{\varepsilon_0},$$

so in general

$$\boxed{E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q_{\text{enc}}(r)}{r^2}}$$

and

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q_{\text{enc}}(r)}{r^2}\,\hat{\mathbf{r}}.}$$

Thus the problem reduces to finding the enclosed charge $Q_{\text{enc}}(r)$ in each case.

(1) Conducting sphere of radius a with total charge Q

For a conductor in electrostatic equilibrium, all excess charge resides on the surface. Therefore:

- for $r < a$, no charge is enclosed;
- for $r > a$, the full charge Q is enclosed.

So for $r < a$,

$$Q_{\text{enc}}(r) = 0,$$

hence

$$\boxed{\mathbf{E}(r) = 0, \qquad r < a.}$$

For $r > a$,

$$Q_{\text{enc}}(r) = Q,$$

so

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,\hat{\mathbf{r}}, \qquad r > a.}$$

At the surface, the field jumps from 0 just inside to

$$E(a^+) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2}$$

just outside.

Therefore the complete result is

$$\boxed{\mathbf{E}(r) = \begin{cases} 0, & 0 \le r < a, \\ \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r^2}\,\hat{\mathbf{r}}, & r > a. \end{cases}}$$

Behavior of $E(r)$: Inside the conductor, the field is identically zero. At $r = a$ there is a discontinuous jump to $Q/(4\pi\varepsilon_0 a^2)$, and for $r > a$ the field falls as $1/r^2$.

(2) Sphere of radius a with uniform volume charge density and total charge Q

Now the charge is spread uniformly throughout the volume. The constant volume charge density is

$$\rho = \frac{Q}{\frac{4}{3}\pi a^3} = \frac{3Q}{4\pi a^3}.$$

Inside the sphere: $r < a$ The charge enclosed by a Gaussian sphere of radius r is

$$Q_{\text{enc}}(r) = \rho\left(\frac{4}{3}\pi r^3\right).$$

Substitute ρ:

$$Q_{\text{enc}}(r) = \frac{3Q}{4\pi a^3} \cdot \frac{4}{3}\pi r^3 = Q\frac{r^3}{a^3}.$$

Then Gauss's law gives

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Qr^3/a^3}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^3}r.$$

Hence

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^3}r\,\hat{\mathbf{r}}, \qquad r < a.}$$

Outside the sphere: $r > a$ The full charge Q is enclosed, so

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,\hat{\mathbf{r}}, \qquad r > a.}$$

Therefore

$$\boxed{\mathbf{E}(r) = \begin{cases} \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{a^3}r\,\hat{\mathbf{r}}, & 0 \le r < a, \\ \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r^2}\,\hat{\mathbf{r}}, & r > a. \end{cases}}$$

At the surface,

$$E(a^-) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2}, \qquad E(a^+) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2},$$

so the field is continuous at $r = a$.

Behavior of $E(r)$: The field starts from zero at the center and rises linearly with r inside the sphere. It reaches its maximum value at the surface,

$$E(a) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2},$$

and then decreases as $1/r^2$ outside.

(3) Sphere of radius a with $\rho(r) = kr^n$, $n > -3$, total charge Q

Now the charge density is spherically symmetric and given by

$$\rho(r) = kr^n, \qquad n > -3.$$

The restriction $n > -3$ is needed so that the total charge near $r = 0$ remains finite.

We first determine the constant k from the condition that the total charge is Q.

Step 1: Determine k The total charge is

$$Q = \int \rho \, d\tau = \int_0^a k r^n (4\pi r^2) \, dr.$$

Thus

$$Q = 4\pi k \int_0^a r^{n+2} \, dr.$$

Since $n > -3$, we have $n + 3 > 0$, so

$$\int_0^a r^{n+2} \, dr = \frac{a^{n+3}}{n+3}.$$

Therefore

$$Q = 4\pi k \frac{a^{n+3}}{n+3},$$

and hence

$$\boxed{k = \frac{Q(n+3)}{4\pi a^{n+3}}.}$$

So the density can be written as

$$\boxed{\rho(r) = \frac{Q(n+3)}{4\pi a^{n+3}} \, r^n.}$$

Step 2: Find the enclosed charge for $r < a$ For $r < a$,

$$Q_{\text{enc}}(r) = \int_0^r \rho(r') \, 4\pi r'^2 \, dr'.$$

Substitute $\rho(r') = k r'^n$:

$$Q_{\text{enc}}(r) = 4\pi k \int_0^r r'^{\,n+2} \, dr' = 4\pi k \frac{r^{n+3}}{n+3}.$$

Now substitute the value of k:

$$Q_{\text{enc}}(r) = 4\pi \left(\frac{Q(n+3)}{4\pi a^{n+3}} \right) \frac{r^{n+3}}{n+3} = Q \frac{r^{n+3}}{a^{n+3}}.$$

Therefore, for $r < a$,

$$E(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q r^{n+3}/a^{n+3}}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^{n+3}} \, r^{n+1}.$$

Hence

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^{n+3}} \, r^{n+1} \, \hat{\mathbf{r}}, \qquad r < a.}$$

Step 3: Outside the sphere, $r > a$ For $r > a$, the full charge Q is enclosed, so

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2} \, \hat{\mathbf{r}}, \qquad r > a.}$$

Thus the full field for the r^n distribution is

$$\boxed{\mathbf{E}(r) = \begin{cases} \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{a^{n+3}} \, r^{n+1} \, \hat{\mathbf{r}}, & 0 \le r < a, \\ \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{r^2} \, \hat{\mathbf{r}}, & r > a. \end{cases}}$$

At the surface,

$$E(a^-) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^2}, \qquad E(a^+) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^2},$$

so the field is continuous at $r = a$, as expected for a finite volume charge density.

Special cases for the sketch

Case $n = -2$ Then

$$\rho(r) = kr^{-2}.$$

Although the density diverges at the origin, the total enclosed charge remains finite because

$$\rho(r)\, 4\pi r^2\, dr \propto dr.$$

Inside the sphere,

$$E(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^{(-2)+3}} r^{(-2)+1} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a} \frac{1}{r}.$$

So

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{ar}\, \hat{\mathbf{r}}, \qquad r < a \quad (n = -2).}$$

Outside,

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}\, \hat{\mathbf{r}}, \qquad r > a.}$$

Behavior: As $r \to 0$, the field diverges like $1/r$. At $r = a$, its value is

$$E(a) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^2},$$

and beyond the sphere it falls as $1/r^2$.

Case $n = +2$ Then

$$\rho(r) = kr^2.$$

Inside the sphere,

$$E(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^5} r^3.$$

So

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{a^5} r^3\, \hat{\mathbf{r}}, \qquad r < a \quad (n = 2).}$$

Outside,

$$\boxed{\mathbf{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}\, \hat{\mathbf{r}}, \qquad r > a.}$$

Behavior: The field starts from zero at the center and rises as r^3, more slowly than linearly near the center, then matches continuously onto the outside Coulomb field at $r = a$.

Final results

$$\boxed{\mathbf{E}_{\text{conducting}}(r) = \begin{cases} 0, & r < a, \\ \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{r^2}\, \hat{\mathbf{r}}, & r > a, \end{cases}}$$

$$\boxed{\mathbf{E}_{\text{uniform}}(r) = \begin{cases} \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{a^3} r\, \hat{\mathbf{r}}, & r < a, \\ \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{r^2}\, \hat{\mathbf{r}}, & r > a, \end{cases}}$$

$$\boxed{\mathbf{E}_{r^n}(r) = \begin{cases} \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{a^{n+3}}\, r^{n+1}\, \hat{\mathbf{r}}, & r < a, \\ \dfrac{1}{4\pi\varepsilon_0} \dfrac{Q}{r^2}\, \hat{\mathbf{r}}, & r > a, \end{cases} \qquad n > -3.}$$

The qualitative sketches are therefore:

- **Conducting sphere:** $E = 0$ for $r < a$, jump at $r = a$, then $1/r^2$ decay.
- **Uniform sphere:** $E \propto r$ for $r < a$, continuous at $r = a$, then $1/r^2$ decay.
- $n = -2$**:** $E \propto 1/r$ for $r < a$, diverges at the origin, continuous at $r = a$, then $1/r^2$ decay.
- $n = +2$**:** $E \propto r^3$ for $r < a$, starts very flat near the origin, continuous at $r = a$, then $1/r^2$ decay.

Problem 1.21

8.1 Problem Statement

A two-dimensional potential problem consists of a unit square area

$$0 \le x \le 1, \qquad 0 \le y \le 1$$

bounded by "surfaces" held at zero potential. Over the entire square there is a uniform charge density of unit strength (per unit length in z).

(a) Apply the variational principle (1.63) for the Poisson equation with the "variational" trial function

$$\Psi(x, y) = A\, x(1 - x)\, y(1 - y)$$

to determine the best value of the constant A. [I use quotation marks around variational because there are no parameters to vary except the overall scale.]

(b) The exact (albeit series) solution for this problem is [see Problems 2.15 and 2.16]

$$4\pi\varepsilon_0\Phi(x, y) = \frac{16}{\pi^3} \sum_{m=0}^{\infty} \frac{\sin[(2m + 1)\pi x]}{(2m + 1)^3} \left[1 - \frac{\cosh[(2m + 1)\pi(y - \frac{1}{2})]}{\cosh[(2m + 1)\pi/2]}\right].$$

For $y = 0.25$ and $y = 0.5$, plot and compare the simple variational solution of part **a** with the exact solution as a function of x.

8.2 Solution

We solve the two-dimensional Poisson equation

$$\nabla^2\Phi(x, y) = -\frac{\rho}{\varepsilon_0},$$

with

$$\rho = \text{constant} = 1,$$

inside the unit square, together with the boundary condition

$$\Phi = 0 \qquad \text{on} \qquad x = 0,\ x = 1,\ y = 0,\ y = 1.$$

The trial function

$$\Psi(x, y) = A\, x(1 - x)\, y(1 - y)$$

already satisfies all boundary conditions, because it vanishes whenever

$$x = 0,\ x = 1,\ y = 0,\ \text{or } y = 1.$$

So the problem is to determine the best value of the single parameter A by means of the variational principle.

Part (a): Variational approximation

For Poisson's equation,

$$\nabla^2\Phi = -\frac{\rho}{\varepsilon_0},$$

with Dirichlet boundary conditions, the appropriate functional is

$$I[\Psi] = \iint \left[\frac{\varepsilon_0}{2}(\nabla\Psi)^2 - \rho\,\Psi\right] dx\,dy.$$

The physical solution makes this functional stationary, and in the present one-parameter trial family we determine A by minimizing $I(A)$.

Since

$$\rho = 1,$$

we have

$$I(A) = \int_0^1\int_0^1 \left[\frac{\varepsilon_0}{2}(\nabla\Psi)^2 - \Psi\right] dx\,dy.$$

We now compute the gradient of

$$\Psi(x,y) = A\,x(1-x)\,y(1-y).$$

First,

$$\frac{\partial\Psi}{\partial x} = A(1-2x)\,y(1-y),$$

and

$$\frac{\partial\Psi}{\partial y} = A\,x(1-x)(1-2y).$$

Therefore

$$(\nabla\Psi)^2 = \left(\frac{\partial\Psi}{\partial x}\right)^2 + \left(\frac{\partial\Psi}{\partial y}\right)^2 = A^2(1-2x)^2y^2(1-y)^2 + A^2x^2(1-x)^2(1-2y)^2.$$

Substituting into the functional,

$$I(A) = \int_0^1\int_0^1 \left[\frac{\varepsilon_0}{2}\left(A^2(1-2x)^2y^2(1-y)^2 + A^2x^2(1-x)^2(1-2y)^2\right) - A\,x(1-x)\,y(1-y)\right] dx\,dy.$$

We now separate the integrals.

First term:

$$\int_0^1\int_0^1 (1-2x)^2y^2(1-y)^2\,dx\,dy = \left(\int_0^1 (1-2x)^2dx\right)\left(\int_0^1 y^2(1-y)^2dy\right).$$

Second term:

$$\int_0^1\int_0^1 x^2(1-x)^2(1-2y)^2\,dx\,dy = \left(\int_0^1 x^2(1-x)^2dx\right)\left(\int_0^1 (1-2y)^2dy\right).$$

Third term:

$$\int_0^1\int_0^1 x(1-x)y(1-y)\,dx\,dy = \left(\int_0^1 x(1-x)\,dx\right)\left(\int_0^1 y(1-y)\,dy\right).$$

So we need the one-dimensional integrals.

First,

$$\int_0^1 (1-2x)^2dx = \int_0^1 (1-4x+4x^2)\,dx = \left[x - 2x^2 + \frac{4}{3}x^3\right]_0^1 = 1 - 2 + \frac{4}{3} = \frac{1}{3}.$$

Next,

$$\int_0^1 x^2(1-x)^2\,dx = \int_0^1 (x^2 - 2x^3 + x^4)\,dx = \left[\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5}\right]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{5} = \frac{10-15+6}{30} = \frac{1}{30}.$$

By symmetry,

$$\int_0^1 y^2(1-y)^2\,dy = \frac{1}{30}, \qquad \int_0^1 (1-2y)^2\,dy = \frac{1}{3}.$$

Also,

$$\int_0^1 x(1-x)\,dx = \int_0^1 (x-x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6},$$

and similarly

$$\int_0^1 y(1-y)\,dy = \frac{1}{6}.$$

Therefore,

$$\int_0^1\int_0^1 (1-2x)^2 y^2 (1-y)^2\,dx\,dy = \frac{1}{3}\cdot\frac{1}{30} = \frac{1}{90},$$

and

$$\int_0^1\int_0^1 x^2(1-x)^2(1-2y)^2\,dx\,dy = \frac{1}{30}\cdot\frac{1}{3} = \frac{1}{90}.$$

Finally,

$$\int_0^1\int_0^1 x(1-x)y(1-y)\,dx\,dy = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}.$$

Hence

$$I(A) = \frac{\varepsilon_0}{2}A^2\left(\frac{1}{90}+\frac{1}{90}\right) - A\left(\frac{1}{36}\right).$$

Since

$$\frac{1}{90}+\frac{1}{90} = \frac{1}{45},$$

we get

$$I(A) = \frac{\varepsilon_0}{2}\frac{A^2}{45} - \frac{A}{36} = \frac{\varepsilon_0 A^2}{90} - \frac{A}{36}.$$

Now minimize with respect to A:

$$\frac{\mathrm{d}I}{\mathrm{d}A} = \frac{2\varepsilon_0 A}{90} - \frac{1}{36} = \frac{\varepsilon_0 A}{45} - \frac{1}{36}.$$

Set this equal to zero:

$$\frac{\varepsilon_0 A}{45} - \frac{1}{36} = 0.$$

So

$$\frac{\varepsilon_0 A}{45} = \frac{1}{36},$$

and therefore

$$A = \frac{45}{36\varepsilon_0} = \frac{5}{4\varepsilon_0}.$$

Thus the best variational approximation is

$$\boxed{\Psi(x,y) = \frac{5}{4\varepsilon_0}\,x(1-x)\,y(1-y).}$$

It is often convenient to write the result in terms of $4\pi\varepsilon_0\Psi$. Multiplying by $4\pi\varepsilon_0$ gives

$$4\pi\varepsilon_0\Psi(x,y) = 4\pi\varepsilon_0\left(\frac{5}{4\varepsilon_0}\right)x(1-x)y(1-y) = 5\pi\,x(1-x)y(1-y).$$

So equivalently,

$$\boxed{4\pi\varepsilon_0\Psi(x,y) = 5\pi\,x(1-x)\,y(1-y).}$$

Check by direct projection of Poisson's equation

It is useful to check that this result is consistent. Compute

$$\nabla^2\Psi = \frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2}.$$

Since

$$\Psi = A\,x(1-x)y(1-y),$$

we have

$$\frac{\partial^2\Psi}{\partial x^2} = -2A\,y(1-y), \qquad \frac{\partial^2\Psi}{\partial y^2} = -2A\,x(1-x).$$

Thus

$$\nabla^2\Psi = -2A\big[x(1-x)+y(1-y)\big].$$

This is not constant, so the trial function cannot satisfy Poisson's equation exactly. The variational method chooses the value of A that gives the best global compromise over the square.

Part (b): Comparison with the exact series solution

The exact solution is

$$4\pi\varepsilon_0\Phi(x,y) = \frac{16}{\pi^2}\sum_{m=0}^{\infty}\frac{\sin[(2m+1)\pi x]}{(2m+1)^3}\left[1-\frac{\cosh[(2m+1)\pi(y-\frac{1}{2})]}{\cosh[(2m+1)\pi/2]}\right].$$

The variational approximation from part **(a)** gives

$$4\pi\varepsilon_0\Psi(x,y) = 5\pi\, x(1-x)\, y(1-y).$$

We now specialize to the two values of y requested.

For $y = 0.5$: Since

$$y(1-y) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4},$$

the variational approximation becomes

$$4\pi\varepsilon_0\Psi(x,0.5) = 5\pi\, x(1-x)\left(\frac{1}{4}\right) = \frac{5\pi}{4}x(1-x).$$

So

$$\boxed{4\pi\varepsilon_0\Psi(x,0.5) = \frac{5\pi}{4}x(1-x).}$$

The exact solution at $y = 0.5$ simplifies because

$$y-\frac{1}{2} = 0, \qquad \cosh(0) = 1.$$

Hence

$$4\pi\varepsilon_0\Phi(x,0.5) = \frac{16}{\pi^2}\sum_{m=0}^{\infty}\frac{\sin[(2m+1)\pi x]}{(2m+1)^3}\left[1-\frac{1}{\cosh[(2m+1)\pi/2]}\right].$$

Therefore

$$\boxed{4\pi\varepsilon_0\Phi(x,0.5) = \frac{16}{\pi^2}\sum_{m=0}^{\infty}\frac{\sin[(2m+1)\pi x]}{(2m+1)^3}\left[1-\frac{1}{\cosh[(2m+1)\pi/2]}\right].}$$

For $y = 0.25$: Now

$$y(1-y) = \frac{1}{4}\cdot\frac{3}{4} = \frac{3}{16},$$

so the variational approximation becomes

$$4\pi\varepsilon_0\Psi(x,0.25) = 5\pi\, x(1-x)\left(\frac{3}{16}\right) = \frac{15\pi}{16}x(1-x).$$

Thus

$$\boxed{4\pi\varepsilon_0\Psi(x,0.25) = \frac{15\pi}{16}x(1-x).}$$

For the exact solution,

$$y-\frac{1}{2} = -\frac{1}{4},$$

and since cosh is even,

$$\cosh\left[(2m+1)\pi\left(-\frac{1}{4}\right)\right] = \cosh\left[(2m+1)\frac{\pi}{4}\right].$$

Hence

$$4\pi\varepsilon_0\Phi(x,0.25) = \frac{16}{\pi^2}\sum_{m=0}^{\infty}\frac{\sin[(2m+1)\pi x]}{(2m+1)^3}\left[1-\frac{\cosh[(2m+1)\pi/4]}{\cosh[(2m+1)\pi/2]}\right].$$

Therefore

$$\boxed{4\pi\varepsilon_0\Phi(x,0.25) = \frac{16}{\pi^2}\sum_{m=0}^{\infty}\frac{\sin[(2m+1)\pi x]}{(2m+1)^3}\left[1-\frac{\cosh[(2m+1)\pi/4]}{\cosh[(2m+1)\pi/2]}\right].}$$

Qualitative comparison

Both the exact and variational solutions vanish at

$$x = 0, \qquad x = 1,$$

and both are symmetric about

$$x = \frac{1}{2}.$$

The variational result is simply a parabola in x for each fixed y:

$$x(1-x).$$

So it gives the correct boundary behavior and the correct symmetry, but it cannot reproduce all the detailed curvature of the exact Fourier-series solution.

At the midpoint $x = 1/2$:

For $y = 0.5$,

$$4\pi\varepsilon_0\Psi\left(\frac{1}{2}, 0.5\right) = \frac{5\pi}{4} \cdot \frac{1}{4} = \frac{5\pi}{16}.$$

For $y = 0.25$,

$$4\pi\varepsilon_0\Psi\left(\frac{1}{2}, 0.25\right) = \frac{15\pi}{16} \cdot \frac{1}{4} = \frac{15\pi}{64}.$$

So the variational approximation predicts the maximum values

$$\boxed{4\pi\varepsilon_0\Psi\left(\frac{1}{2}, 0.5\right) = \frac{5\pi}{16}, \qquad 4\pi\varepsilon_0\Psi\left(\frac{1}{2}, 0.25\right) = \frac{15\pi}{64}.}$$

These provide convenient reference values for comparison with the exact series.

Thus the final variational solution is

$$\boxed{\Psi(x,y) = \frac{5}{4\varepsilon_0}\, x(1-x)\, y(1-y),}$$

or equivalently

$$\boxed{4\pi\varepsilon_0\Psi(x,y) = 5\pi\, x(1-x)\, y(1-y).}$$

And for the requested slices:

$$\boxed{4\pi\varepsilon_0\Psi(x, 0.5) = \frac{5\pi}{4}x(1-x), \qquad 4\pi\varepsilon_0\Psi(x, 0.25) = \frac{15\pi}{16}x(1-x).}$$

Problem 1.5

9.1 Problem Statement

The time-averaged potential of a neutral hydrogen atom is given by

$$\Phi(r) = \frac{q}{4\pi\varepsilon_0}\frac{e^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right),$$

where q is the magnitude of the electronic charge, and $\alpha^{-1} = a_0/2$, a_0 being the Bohr radius. Find the distribution of charge (both continuous and discrete) that will give this potential and interpret your result physically.

9.2 Solution

We are given a spherically symmetric potential,

$$\Phi(r) = \frac{q}{4\pi\varepsilon_0}\frac{e^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right).$$

Since the potential depends only on r, the corresponding charge distribution must also be spherically symmetric.

The relation between potential and charge density is Poisson's equation:

$$\nabla^2\Phi = -\frac{\rho}{\varepsilon_0}.$$

Therefore our task is to compute $\nabla^2\Phi$, identify any delta-function piece at the origin, and then read off the continuous and discrete parts of the charge distribution.

Step 1: Separate the potential into simpler pieces

Write

$$\Phi(r) = \frac{q}{4\pi\varepsilon_0}\left(\frac{e^{-\alpha r}}{r} + \frac{\alpha}{2}e^{-\alpha r}\right).$$

So it is enough to know the Laplacian of the two functions

$$\frac{e^{-\alpha r}}{r} \qquad \text{and} \qquad e^{-\alpha r}.$$

For a spherically symmetric function $f(r)$,

$$\nabla^2 f = \frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}f}{\mathrm{d}r}\right), \qquad r \neq 0.$$

However, for functions behaving like $1/r$ near the origin, one must also include the distributional identity

$$\nabla^2\left(\frac{1}{r}\right) = -4\pi\,\delta^3(\mathbf{r}).$$

More generally,

$$\boxed{\nabla^2\left(\frac{e^{-\alpha r}}{r}\right) = \alpha^2\frac{e^{-\alpha r}}{r} - 4\pi\,\delta^3(\mathbf{r})}$$

which is the Yukawa-potential identity.

We will also compute

$$\nabla^2(e^{-\alpha r}).$$

Step 2: Compute $\nabla^2(e^{-\alpha r})$

Let

$$f(r) = e^{-\alpha r}.$$

Then

$$\frac{\mathrm{d}f}{\mathrm{d}r} = -\alpha e^{-\alpha r}.$$

Hence

$$r^2\frac{\mathrm{d}f}{\mathrm{d}r} = -\alpha r^2 e^{-\alpha r}.$$

Differentiate:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}f}{\mathrm{d}r}\right) = \frac{\mathrm{d}}{\mathrm{d}r}\left(-\alpha r^2 e^{-\alpha r}\right).$$

Using the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 e^{-\alpha r}\right) = 2re^{-\alpha r} - \alpha r^2 e^{-\alpha r},$$

so

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(-\alpha r^2 e^{-\alpha r}\right) = -\alpha\left(2re^{-\alpha r} - \alpha r^2 e^{-\alpha r}\right).$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}f}{\mathrm{d}r}\right) = -2\alpha re^{-\alpha r} + \alpha^2 r^2 e^{-\alpha r}.$$

Divide by r^2:

$$\nabla^2(e^{-\alpha r}) = -\frac{2\alpha}{r}e^{-\alpha r} + \alpha^2 e^{-\alpha r}, \qquad r \neq 0.$$

Therefore

$$\boxed{\nabla^2(e^{-\alpha r}) = \alpha^2 e^{-\alpha r} - \frac{2\alpha}{r}e^{-\alpha r} \qquad (r \neq 0).}$$

Step 3: Compute the Laplacian of the full potential

Now

$$\Phi(r) = \frac{q}{4\pi\varepsilon_0}\left(\frac{e^{-\alpha r}}{r} + \frac{\alpha}{2}e^{-\alpha r}\right),$$

so

$$\nabla^2\Phi = \frac{q}{4\pi\varepsilon_0}\left[\nabla^2\left(\frac{e^{-\alpha r}}{r}\right) + \frac{\alpha}{2}\nabla^2(e^{-\alpha r})\right].$$

Substitute the identities above:

$$\nabla^2\Phi = \frac{q}{4\pi\varepsilon_0}\left[\alpha^2\frac{e^{-\alpha r}}{r} - 4\pi\delta^3(\mathbf{r}) + \frac{\alpha}{2}\left(\alpha^2 e^{-\alpha r} - \frac{2\alpha}{r}e^{-\alpha r}\right)\right].$$

Distribute the factor $\alpha/2$:

$$\nabla^2\Phi = \frac{q}{4\pi\varepsilon_0}\left[\alpha^2\frac{e^{-\alpha r}}{r} - 4\pi\delta^3(\mathbf{r}) + \frac{\alpha^3}{2}e^{-\alpha r} - \alpha^2\frac{e^{-\alpha r}}{r}\right].$$

The two $1/r$ terms cancel:

$$\alpha^2\frac{e^{-\alpha r}}{r} - \alpha^2\frac{e^{-\alpha r}}{r} = 0.$$

So the result simplifies drastically:

$$\nabla^2\Phi = \frac{q}{4\pi\varepsilon_0}\left[-4\pi\delta^3(\mathbf{r}) + \frac{\alpha^3}{2}e^{-\alpha r}\right].$$

Therefore, from Poisson's equation,

$$\rho(\mathbf{r}) = -\varepsilon_0\nabla^2\Phi,$$

we obtain

$$\rho(\mathbf{r}) = -\varepsilon_0\frac{q}{4\pi\varepsilon_0}\left[-4\pi\delta^3(\mathbf{r}) + \frac{\alpha^3}{2}e^{-\alpha r}\right].$$

Cancel ε_0:

$$\rho(\mathbf{r}) = -\frac{q}{4\pi}\left[-4\pi\delta^3(\mathbf{r}) + \frac{\alpha^3}{2}e^{-\alpha r}\right].$$

Distribute the minus sign:

$$\rho(\mathbf{r}) = q\,\delta^3(\mathbf{r}) - \frac{q\alpha^3}{8\pi}e^{-\alpha r}.$$

Hence the charge distribution is

$$\boxed{\rho(\mathbf{r}) = q\,\delta^3(\mathbf{r}) - \frac{q\alpha^3}{8\pi}e^{-\alpha r}.}$$

This already displays the answer in the required form: there is a discrete point charge at the origin, and a continuous negative charge cloud surrounding it.

Step 4: Identify the discrete and continuous parts

The discrete part is

$$\boxed{\rho_{\text{disc}}(\mathbf{r}) = q\,\delta^3(\mathbf{r}).}$$

This is a point charge $+q$ at the origin, which is the proton.

The continuous part is

$$\boxed{\rho_{\text{cont}}(\mathbf{r}) = -\frac{q\alpha^3}{8\pi}e^{-\alpha r}.}$$

This is the spherically symmetric electronic charge cloud.

Step 5: Check that the total charge is zero

Because the hydrogen atom is neutral, the total charge should vanish. Let us verify that.

The total discrete charge is simply

$$Q_{\text{disc}} = q.$$

The total continuous charge is

$$Q_{\text{cont}} = \iiint \rho_{\text{cont}}\, d\tau = \int_0^\infty \left(-\frac{q\alpha^3}{8\pi} e^{-\alpha r}\right) 4\pi r^2\, dr.$$

Thus

$$Q_{\text{cont}} = -\frac{q\alpha^3}{2} \int_0^\infty r^2 e^{-\alpha r}\, dr.$$

Now use the standard integral

$$\int_0^\infty r^2 e^{-\alpha r}\, dr = \frac{2}{\alpha^3}.$$

Therefore

$$Q_{\text{cont}} = -\frac{q\alpha^3}{2} \cdot \frac{2}{\alpha^3} = -q.$$

So

$$Q_{\text{tot}} = Q_{\text{disc}} + Q_{\text{cont}} = q - q = 0,$$

as required.

Step 6: Relation to the hydrogen ground-state charge cloud

Since

$$\alpha^{-1} = \frac{a_0}{2}, \qquad \text{so} \qquad \alpha = \frac{2}{a_0},$$

the continuous charge density becomes

$$\rho_{\text{cont}}(r) = -\frac{q}{8\pi}\left(\frac{2}{a_0}\right)^3 e^{-2r/a_0} = -\frac{q}{\pi a_0^3} e^{-2r/a_0}.$$

Hence

$$\boxed{\rho(\mathbf{r}) = q\,\delta^3(\mathbf{r}) - \frac{q}{\pi a_0^3} e^{-2r/a_0}.}$$

This is exactly the familiar charge distribution for the time-averaged hydrogen atom in the ground state: a point proton at the origin plus the smeared-out electron charge density

$$\rho_e(r) = -\frac{q}{\pi a_0^3} e^{-2r/a_0}.$$

Physical interpretation

The potential is not simply the Coulomb potential of a point charge because the atom is not a bare proton. Instead, it consists of:

- a *positive point charge* $+q$ at the origin, representing the proton;
- a *negative continuous charge cloud* whose density decreases exponentially with r, representing the electron probability cloud in the $1s$ state.

Near the origin, the potential behaves approximately like

$$\Phi(r) \sim \frac{q}{4\pi\varepsilon_0} \frac{1}{r},$$

because very close to the origin the proton dominates.

Far from the atom,

$$e^{-\alpha r} \to 0,$$

so the potential falls off exponentially to zero rather than as $1/r$. This reflects the fact that the total charge is zero: at large distances the positive nuclear charge is screened by the negative electron cloud.

Therefore the given potential corresponds physically to a neutral hydrogen atom whose time-averaged charge distribution is

$$\boxed{\rho(\mathbf{r}) = q\,\delta^3(\mathbf{r}) - \frac{q\alpha^3}{8\pi}e^{-\alpha r}}$$

or, using $\alpha = 2/a_0$,

$$\boxed{\rho(\mathbf{r}) = q\,\delta^3(\mathbf{r}) - \frac{q}{\pi a_0^3}e^{-2r/a_0}.}$$

Problem 1.20

10.1 Problem Statement

In estimating the capacitance of a given configuration of conductors, comparison with known capacitances is often helpful. Consider two configurations of n conductors in which the $(n-1)$ conductors held at zero potential are the same, but the one conductor whose capacitance we wish to know is different. In particular, let the conductor in one configuration have a closed surface S_1 and in the other configu- ration have surface S_1', with S_1' totally inside S_1.

(a) Use the extremum principle of Section 1.12 and the variational principle of Problem 1.17 to prove that the capacitance C' of the conductor with surface S_1' is less than or equal to the capacitance C of the conductor with surface S_1 that encloses S_1'.

(b) Set upper and lower limits for the capacitance of a conducting cube of side a. Compare your limits and also their average with the numerical value,

$$C \simeq 0.655(4\pi\varepsilon_0 a).$$

(c) By how much do you estimate the capacitance per unit length of the two-wire system of Problem 1.7 will change (larger? smaller?) if one of the wires is replaced by a wire of square cross section whose side is equal to its diameter?

10.2 Solution

We consider electrostatics in vacuum. If one conductor is held at potential V and all other conductors are grounded, then the capacitance is defined by

$$C = \frac{Q}{V},$$

where Q is the total charge on the conductor at potential V.

It is often convenient to choose

$$V = 1.$$

Then the capacitance is numerically equal to the total charge,

$$C = Q.$$

There is also a very useful energy formula. The electrostatic energy is

$$U = \frac{\varepsilon_0}{2}\int_{\text{all space}} |\mathbf{E}|^2\,d\tau.$$

Since $\mathbf{E} = -\nabla\Phi$, this becomes

$$U = \frac{\varepsilon_0}{2}\int |\nabla\Phi|^2\,d\tau.$$

For a capacitor with one conductor at potential V and the others at 0,

$$U = \frac{1}{2}CV^2.$$

If we choose $V = 1$, then

$$\boxed{C = \varepsilon_0\int |\nabla\Phi|^2\,d\tau.}$$

So when the conductor at nonzero potential is fixed at 1, the capacitance is the minimum value of the functional

$$\mathcal{I}[\Phi] = \varepsilon_0\int |\nabla\Phi|^2\,d\tau$$

among all trial functions Φ satisfying the required boundary values. This is the variational principle we need.

(a) Proof that $C' \leq C$ when S_1' lies inside S_1

Let configuration A be the one with conductor surface S_1, and let its exact potential be $\Phi(\mathbf{r})$. Let configuration B be the one with conductor surface S_1', and let its exact potential be $\Phi'(\mathbf{r})$. In both cases all the other conductors are the same and are grounded.

We set the conductor of interest at unit potential:

$$\Phi = 1 \quad \text{on } S_1, \qquad \Phi' = 1 \quad \text{on } S_1',$$

and

$$\Phi = \Phi' = 0 \quad \text{on all the other conductor surfaces.}$$

Then

$$C = \varepsilon_0 \int |\nabla \Phi|^2 \, d\tau, \qquad C' = \varepsilon_0 \int |\nabla \Phi'|^2 \, d\tau.$$

We want to show that

$$C' \leq C.$$

Step 1: Use the potential of the larger conductor as a trial function for the smaller one. Because S_1' lies wholly inside S_1, the region exterior to S_1 is contained in the region exterior to S_1'. The exact potential Φ for the larger conductor satisfies

$$\Phi = 1 \quad \text{on } S_1, \qquad \Phi = 0 \quad \text{on the grounded conductors.}$$

Now consider the same function Φ in the larger region exterior to S_1'. Since S_1' lies inside S_1, the surface S_1' is not in general an equipotential of Φ. So Φ cannot be used directly as an admissible trial function for configuration B.

However, by the extremum principle of electrostatics, in a charge-free region bounded by conductors, the exact potential takes values only between the extreme boundary values. Since the boundary values here are 0 and 1,

$$0 \leq \Phi(\mathbf{r}) \leq 1$$

everywhere in the field region.

In particular, on the inner surface S_1',

$$0 \leq \Phi \leq 1.$$

Now define a modified trial function

$$\widetilde{\Phi}(\mathbf{r}) = \begin{cases} 1, & \mathbf{r} \text{ inside } S_1, \\ \Phi(\mathbf{r}), & \mathbf{r} \text{ outside } S_1. \end{cases}$$

Since S_1' lies entirely inside S_1, this trial function satisfies

$$\widetilde{\Phi} = 1 \quad \text{on } S_1',$$

and still

$$\widetilde{\Phi} = 0 \quad \text{on all the grounded conductors.}$$

Therefore $\widetilde{\Phi}$ is an admissible trial function for configuration B.

Step 2: Compare the corresponding energies. Inside S_1, the trial function $\widetilde{\Phi}$ is constant, so

$$\nabla \widetilde{\Phi} = 0.$$

Outside S_1, it agrees with Φ, so

$$\nabla \widetilde{\Phi} = \nabla \Phi.$$

Hence

$$\int |\nabla \widetilde{\Phi}|^2 \, d\tau = \int_{\text{outside } S_1} |\nabla \Phi|^2 \, d\tau.$$

But for the original configuration with surface S_1, the electric field exists only outside S_1, so this is exactly

$$\int |\nabla \widetilde{\Phi}|^2 \, d\tau = \int |\nabla \Phi|^2 \, d\tau.$$

Therefore the trial function $\widetilde{\Phi}$ gives the value

$$\varepsilon_0 \int |\nabla \widetilde{\Phi}|^2 \, d\tau = C.$$

Now use the variational principle for configuration B: the exact potential Φ' gives the *minimum* of the functional among all admissible trial functions, so

$$C' = \varepsilon_0 \int |\nabla \Phi'|^2 \, d\tau \leq \varepsilon_0 \int |\nabla \widetilde{\Phi}|^2 \, d\tau = C.$$

Thus

$$\boxed{C' \leq C.}$$

This proves that enlarging the conductor while leaving all the grounded conductors fixed can only increase the capacitance.

Physical meaning. A larger conductor held at the same potential can store more charge because its electric field can spread more effectively through the surrounding region. Equivalently, when the shape expands outward toward the grounded conductors, the same potential difference can be maintained with a larger total charge.

(b) Bounds for the capacitance of a conducting cube of side a

Let C_{cube} be the capacitance of an isolated conducting cube of side a.

We compare the cube with spheres:

- the largest sphere completely inside the cube;
- the smallest sphere completely enclosing the cube.

Since for an isolated sphere of radius R,

$$C_{\text{sphere}} = 4\pi\varepsilon_0 R,$$

these comparisons immediately give bounds.

Lower bound: inscribed sphere. The largest sphere inside the cube has radius

$$R_{\text{in}} = \frac{a}{2}.$$

Because the cube encloses this sphere, part **(a)** implies

$$C_{\text{cube}} \geq C_{\text{in}}.$$

Therefore

$$C_{\text{cube}} \geq 4\pi\varepsilon_0 \left(\frac{a}{2}\right) = 2\pi\varepsilon_0 a.$$

So

$$\boxed{C_{\text{cube}} \geq 2\pi\varepsilon_0 a.}$$

Upper bound: circumscribed sphere. The smallest sphere containing the cube is the one passing through the cube's corners. Its radius is half the body diagonal:

$$R_{\text{out}} = \frac{\sqrt{3}\, a}{2}.$$

This sphere encloses the cube, so by part **(a)**,

$$C_{\text{cube}} \leq C_{\text{out}}.$$

Hence

$$C_{\text{cube}} \leq 4\pi\varepsilon_0 \left(\frac{\sqrt{3}\, a}{2}\right) = 2\sqrt{3}\, \pi\varepsilon_0 a.$$

So

$$\boxed{C_{\text{cube}} \leq 2\sqrt{3}\, \pi\varepsilon_0 a.}$$

Therefore the bounds are

$$\boxed{2\pi\varepsilon_0 a \leq C_{\text{cube}} \leq 2\sqrt{3}\, \pi\varepsilon_0 a.}$$

If we factor out $4\pi\varepsilon_0 a$, this becomes

$$\boxed{\frac{1}{2} \leq \frac{C_{\text{cube}}}{4\pi\varepsilon_0 a} \leq \frac{\sqrt{3}}{2}.}$$

Comparison with the numerical value. The quoted numerical value is

$$C_{\text{num}} \simeq 0.655(4\pi\varepsilon_0 a).$$

Our bounds correspond to

$$0.500(4\pi\varepsilon_0 a) \leq C_{\text{cube}} \leq 0.866(4\pi\varepsilon_0 a).$$

So the numerical value indeed lies between them:

$$0.500 < 0.655 < 0.866.$$

Average of the bounds. The arithmetic average of the lower and upper bounds is

$$C_{\text{avg}} = \frac{1}{2}\left(2\pi\varepsilon_0 a + 2\sqrt{3}\,\pi\varepsilon_0 a\right) = \pi\varepsilon_0 a(1+\sqrt{3}).$$

Factoring out $4\pi\varepsilon_0 a$,

$$C_{\text{avg}} = \frac{1+\sqrt{3}}{4}(4\pi\varepsilon_0 a).$$

Numerically,

$$\frac{1+\sqrt{3}}{4} \approx \frac{1+1.732}{4} \approx 0.683.$$

Thus

$$\boxed{C_{\text{avg}} \approx 0.683(4\pi\varepsilon_0 a).}$$

Compared with the quoted value $0.655(4\pi\varepsilon_0 a)$, the average is somewhat high, but still reasonably close:

$$0.683 - 0.655 = 0.028.$$

(c) Effect on the capacitance per unit length of the two-wire system

Now consider the two-wire system of Problem 1.7. One wire is replaced by a wire of square cross section whose side equals the diameter of the original circular wire.

Let the original wire have radius b. Then its diameter is

$$2b.$$

The new square has side

$$2b.$$

A circle of radius b is exactly inscribed in a square of side $2b$, so the new square cross section completely encloses the original circular cross section.

Therefore, with the other wire unchanged, the new conductor surface encloses the old one. By part **(a)**, the capacitance per unit length must increase:

$$\boxed{C' > C.}$$

So the change is:

$$\boxed{\text{the capacitance per unit length becomes larger.}}$$

Physical explanation. Replacing the circular wire by the square wire makes that conductor effectively larger. Its surface extends farther outward, especially near the corners, so at the same potential difference it can hold more charge. Equivalently, the electric field is concentrated more strongly near the corners, which increases the total charge stored per unit potential difference. Hence the capacitance increases.

Final answers

$$\boxed{C' \le C \quad \text{if } S_1' \text{ lies entirely inside } S_1.}$$

$$\boxed{2\pi\varepsilon_0 a \le C_{\text{cube}} \le 2\sqrt{3}\,\pi\varepsilon_0 a}$$

or equivalently

$$\boxed{0.500(4\pi\varepsilon_0 a) \le C_{\text{cube}} \le 0.866(4\pi\varepsilon_0 a).}$$

$$\boxed{C_{\text{avg}} = \frac{1+\sqrt{3}}{4}(4\pi\varepsilon_0 a) \approx 0.683(4\pi\varepsilon_0 a).}$$

Replacing one round wire by a square wire of side equal to its diameter makes the capacitance per unit length larger.

Problem 1.6

11.1 Problem Statement

A simple capacitor is a device formed by two insulated conductors adjacent to each other. If equal and opposite charges are placed on the conductors, there will be a certain difference of potential between them. The ratio of the magnitude of the charge on one conductor to the magnitude of the potential difference is called the capacitance (in SI units it is measured in farads). Using Gauss's law, calculate the capacitance of

(a) two large, flat, conducting sheets of area A, separated by a small distance d;

(b) two concentric conducting spheres with radii a, b ($b > a$);

(c) two concentric conducting cylinders of length L, large compared to their radii a, b ($b > a$).

(d) What is the inner diameter of the outer conductor in an air-filled coaxial cable whose center conductor is a cylindrical wire of diameter 1 mm and whose capacitance is 3×10^{-11} F/m? 3×10^{-12} F/m?

11.2 Solution

The capacitance is defined by

$$C = \frac{Q}{V},$$

where Q is the magnitude of the charge on either conductor and V is the potential difference between them.

In each case we first determine the electric field using Gauss's law, then compute the potential difference from

$$V = \int \mathbf{E} \cdot d\mathbf{l},$$

and finally form the ratio Q/V.

(a) Two large parallel conducting sheets

Consider two large flat conducting sheets of area A, carrying charges $+Q$ and $-Q$, separated by a small distance d. Because the sheets are large compared with the separation, edge effects are neglected, so the field is essentially uniform between the sheets.

Let the surface charge density be

$$\sigma = \frac{Q}{A}.$$

A single infinite sheet with surface charge density σ produces an electric field of magnitude

$$E_{\text{sheet}} = \frac{\sigma}{2\varepsilon_0}$$

on each side. For two oppositely charged sheets, the fields between the plates add, while outside they cancel. Therefore the field between the sheets is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}.$$

Since the field is uniform, the potential difference is

$$V = Ed = \frac{Q}{\varepsilon_0 A} d.$$

Hence the capacitance is

$$C = \frac{Q}{V} = \frac{Q}{Qd/(\varepsilon_0 A)} = \frac{\varepsilon_0 A}{d}.$$

Therefore

$$\boxed{C = \frac{\varepsilon_0 A}{d}.}$$

Physical interpretation. A larger plate area allows more charge to be stored for the same potential difference, so C increases with A. Increasing the separation weakens the electric interaction between the plates and increases the required voltage for a given charge, so C decreases with d.

(b) Two concentric conducting spheres

Now consider two concentric conducting spheres of radii a and b, with $b > a$. Let the inner sphere carry charge $+Q$ and the outer sphere carry charge $-Q$.

Because of spherical symmetry, the electric field in the region

$$a < r < b$$

must be radial:

$$\mathbf{E}(r) = E(r)\,\hat{\mathbf{r}}.$$

Choose a spherical Gaussian surface of radius r with

$$a < r < b.$$

Gauss's law gives

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

Since the Gaussian surface encloses only the charge $+Q$ on the inner sphere,

$$Q_{\text{enc}} = Q.$$

Also, since $E(r)$ is constant on the sphere and radial,

$$\oint \mathbf{E} \cdot d\mathbf{a} = E(r)\,4\pi r^2.$$

Therefore

$$E(r)\,4\pi r^2 = \frac{Q}{\varepsilon_0},$$

so

$$\boxed{E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}, \qquad a < r < b.}$$

Now compute the potential difference between the two conductors:

$$V = \Phi(a) - \Phi(b) = \int_a^b E(r)\,dr.$$

Thus

$$V = \int_a^b \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,dr = \frac{Q}{4\pi\varepsilon_0}\int_a^b r^{-2}\,dr.$$

Since

$$\int r^{-2}\,dr = -\frac{1}{r},$$

we obtain

$$V = \frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_a^b = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right).$$

Hence

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)}.$$

Therefore

$$C = \frac{4\pi\varepsilon_0}{\frac{1}{a} - \frac{1}{b}} = 4\pi\varepsilon_0 \frac{ab}{b-a}.$$

So the capacitance is

$$\boxed{C = 4\pi\varepsilon_0 \frac{ab}{b-a}.}$$

Checks. If $b \to \infty$, the outer sphere goes infinitely far away, and we recover the capacitance of an isolated sphere of radius a:

$$C \to 4\pi\varepsilon_0 a.$$

(c) Two concentric conducting cylinders

Consider two concentric conducting cylinders of radii a and b, length L, with

$$b > a,$$

and assume

$$L \gg a, b,$$

so edge effects can be neglected.

Let the inner cylinder carry charge $+Q$ and the outer cylinder $-Q$. By cylindrical symmetry, the electric field in the region

$$a < r < b$$

is radial and depends only on r:

$$\mathbf{E}(r) = E(r)\,\hat{\mathbf{r}}.$$

Take as Gaussian surface a coaxial cylinder of radius r and length L, with

$$a < r < b.$$

Gauss's law gives

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

The enclosed charge is Q, and the flux is through the curved surface only:

$$\oint \mathbf{E} \cdot d\mathbf{a} = E(r)(2\pi r L).$$

Hence

$$E(r)(2\pi r L) = \frac{Q}{\varepsilon_0},$$

so

$$\boxed{E(r) = \frac{Q}{2\pi\varepsilon_0 L}\frac{1}{r}, \qquad a < r < b.}$$

Now compute the potential difference:

$$V = \Phi(a) - \Phi(b) = \int_a^b E(r)\,dr.$$

Thus

$$V = \int_a^b \frac{Q}{2\pi\varepsilon_0 L}\frac{1}{r}\,dr = \frac{Q}{2\pi\varepsilon_0 L}\int_a^b \frac{dr}{r}.$$

Since

$$\int_a^b \frac{dr}{r} = \ln\frac{b}{a},$$

we obtain

$$V = \frac{Q}{2\pi\varepsilon_0 L}\ln\frac{b}{a}.$$

Therefore

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{2\pi\varepsilon_0 L}\ln(b/a)} = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}.$$

So the capacitance is

$$\boxed{C = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}.}$$

The capacitance per unit length is therefore

$$\boxed{\frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln(b/a)}.}$$

(d) Coaxial cable

For an air-filled coaxial cable,

$$\frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln(b/a)}.$$

The center conductor diameter is

$$1 \text{ mm},$$

so its radius is

$$a = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}.$$

We solve for b from

$$\ln\frac{b}{a} = \frac{2\pi\varepsilon_0}{C/L}.$$

Case 1: $C/L = 3 \times 10^{-11}$ F/m Using

$$\varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m},$$

we have

$$2\pi\varepsilon_0 \approx 2\pi(8.854 \times 10^{-12}) \approx 5.56 \times 10^{-11} \text{ F/m}.$$

Therefore

$$\ln\frac{b}{a} = \frac{5.56 \times 10^{-11}}{3 \times 10^{-11}} \approx 1.85.$$

So

$$\frac{b}{a} = e^{1.85} \approx 6.37.$$

Hence

$$b \approx 6.37\, a = 6.37(0.5 \text{ mm}) \approx 3.19 \text{ mm}.$$

Therefore the inner diameter of the outer conductor is

$$2b \approx 6.37 \text{ mm}.$$

So for this case,

$$\boxed{\text{inner diameter} \approx 6.4 \text{ mm}.}$$

Case 2: $C/L = 3 \times 10^{-12}$ F/m Now

$$\ln\frac{b}{a} = \frac{5.56 \times 10^{-11}}{3 \times 10^{-12}} \approx 18.5.$$

Thus

$$\frac{b}{a} = e^{18.5} \approx 1.08 \times 10^{8}.$$

So

$$b \approx (1.08 \times 10^{8})(0.5 \text{ mm}) \approx 5.4 \times 10^{4} \text{ m}.$$

Therefore the inner diameter is

$$2b \approx 1.1 \times 10^{5} \text{ m}.$$

Hence

$$\boxed{\text{inner diameter} \approx 1.1 \times 10^{5} \text{ m} \approx 110 \text{ km}.}$$

Physical comment. This enormous value in the second case is reasonable mathematically: a very small capacitance per unit length means the conductors must be extremely weakly coupled, which requires the outer conductor to be extraordinarily far away.

Final answers

$$\boxed{C_{\text{parallel plates}} = \frac{\varepsilon_0 A}{d}}$$

$$\boxed{C_{\text{spherical}} = 4\pi\varepsilon_0 \frac{ab}{b-a}}$$

$$\boxed{C_{\text{cylindrical}} = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}} \qquad \text{and} \qquad \boxed{\frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln(b/a)}}$$

For the coaxial cable:

$$\boxed{2b \approx 6.4 \text{ mm} \quad \text{for} \quad \frac{C}{L} = 3 \times 10^{-11} \text{ F/m}}$$

and

$$\boxed{2b \approx 1.1 \times 10^5 \text{ m} \quad \text{for} \quad \frac{C}{L} = 3 \times 10^{-12} \text{ F/m.}}$$

Problem 1.19

12.1 Problem Statement

For the cylindrical capacitor of Problem 1.6c, evaluate the variational upper bound of Problem 1.17b with the naive trial function

$$\Psi_1(\rho) = \frac{b-\rho}{b-a}.$$

Compare the variational result with the exact result for

$$\frac{b}{a} = 1.5,\ 2,\ 3.$$

Explain the trend of your results in terms of the functional form of Ψ_1. An improved trial function is treated by Collin (pp. 275–277).

12.2 Solution

We consider the coaxial cylindrical capacitor of Problem 1.6c. The inner conductor has radius a, the outer conductor has radius b, and the common length is L, with

$$b > a.$$

The exact capacitance is

$$\boxed{C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}.}$$

The purpose of this problem is to evaluate the *variational upper bound* using the trial potential

$$\Psi_1(\rho) = \frac{b-\rho}{b-a}.$$

This trial function satisfies the correct boundary conditions:

$$\Psi_1(a) = 1, \qquad \Psi_1(b) = 0.$$

So it is an admissible trial function for the Dirichlet variational principle.

1. Variational principle for the capacitance

For a capacitor with one conductor held at unit potential and the other at zero potential, the exact electrostatic energy is

$$U = \frac{1}{2}C.$$

On the other hand, for any admissible potential function Ψ satisfying the same boundary values,

$$U[\Psi] = \frac{\varepsilon_0}{2}\int_\tau |\nabla\Psi|^2\, d\tau.$$

The exact solution minimizes this functional, so

$$\frac{1}{2}C \le \frac{\varepsilon_0}{2}\int_\tau |\nabla\Psi|^2\, d\tau.$$

Multiplying by 2,

$$\boxed{C \le \varepsilon_0 \int_\tau |\nabla\Psi|^2\, d\tau.}$$

Thus any admissible trial potential gives an *upper bound* on the capacitance.

So for the trial function Ψ_1,

$$C_{\text{var}} = \varepsilon_0 \int_\tau |\nabla\Psi_1|^2\, d\tau$$

is the desired upper bound.

2. Evaluate the functional for the trial function

The trial function depends only on the cylindrical radial coordinate ρ, so in cylindrical coordinates

$$\nabla\Psi_1 = \hat{\boldsymbol{\rho}}\,\frac{\mathrm{d}\Psi_1}{\mathrm{d}\rho}.$$

Now

$$\Psi_1(\rho) = \frac{b-\rho}{b-a},$$

so

$$\frac{\mathrm{d}\Psi_1}{\mathrm{d}\rho} = -\frac{1}{b-a}.$$

Therefore

$$|\nabla\Psi_1|^2 = \left(\frac{\mathrm{d}\Psi_1}{\mathrm{d}\rho}\right)^2 = \frac{1}{(b-a)^2}.$$

The field region is the annular cylindrical volume

$$a \le \rho \le b, \qquad 0 \le \phi \le 2\pi, \qquad 0 \le z \le L.$$

The volume element is

$$d\tau = \rho\, d\rho\, d\phi\, dz.$$

Hence

$$C_{\text{var}} = \varepsilon_0 \int_0^L \int_0^{2\pi} \int_a^b \frac{1}{(b-a)^2}\,\rho\, d\rho\, d\phi\, dz.$$

Since the integrand is constant except for the factor ρ, the integrals separate:

$$C_{\text{var}} = \varepsilon_0 \frac{1}{(b-a)^2}\left(\int_0^L dz\right)\left(\int_0^{2\pi} d\phi\right)\left(\int_a^b \rho\, d\rho\right).$$

Compute each factor:

$$\int_0^L dz = L, \qquad \int_0^{2\pi} d\phi = 2\pi, \qquad \int_a^b \rho\, d\rho = \frac{b^2-a^2}{2}.$$

Therefore

$$C_{\text{var}} = \varepsilon_0 \frac{1}{(b-a)^2}\, L\,(2\pi)\,\frac{b^2-a^2}{2}.$$

The factor 2 cancels:

$$C_{\text{var}} = \pi\varepsilon_0 L \frac{b^2 - a^2}{(b-a)^2}.$$

Since

$$b^2 - a^2 = (b-a)(b+a),$$

we obtain

$$\boxed{C_{\text{var}} = \pi\varepsilon_0 L \frac{a+b}{b-a}.}$$

This is the variational upper bound obtained from the naive trial function.

3. Compare with the exact result

The exact capacitance is

$$C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}.$$

So the ratio of the variational estimate to the exact value is

$$\frac{C_{\text{var}}}{C_{\text{exact}}} = \frac{\pi\varepsilon_0 L \dfrac{a+b}{b-a}}{\dfrac{2\pi\varepsilon_0 L}{\ln(b/a)}}.$$

Canceling the common factors $\pi\varepsilon_0 L$ gives

$$\boxed{\frac{C_{\text{var}}}{C_{\text{exact}}} = \frac{a+b}{2(b-a)} \ln\frac{b}{a}.}$$

It is convenient to define

$$\beta = \frac{b}{a}.$$

Then

$$b = \beta a,$$

and

$$\frac{a+b}{b-a} = \frac{a+\beta a}{\beta a - a} = \frac{1+\beta}{\beta - 1}.$$

Hence

$$\boxed{C_{\text{var}} = \pi\varepsilon_0 L \frac{\beta+1}{\beta-1},}$$

$$\boxed{C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln\beta},}$$

and

$$\boxed{\frac{C_{\text{var}}}{C_{\text{exact}}} = \frac{\beta+1}{2(\beta-1)} \ln\beta.}$$

We now evaluate these for the requested values.

4. Case $b/a = 1.5$

Here

$$\beta = 1.5.$$

The exact result is

$$C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln(1.5)}.$$

Using

$$\ln(1.5) \approx 0.405465,$$

we get

$$C_{\text{exact}} \approx 15.496\,\varepsilon_0 L.$$

Equivalently,

$$\boxed{C_{\text{exact}} \approx 15.496\,\varepsilon_0 L.}$$

The variational estimate is

$$C_{\text{var}} = \pi\varepsilon_0 L\,\frac{1+1.5}{1.5-1} = \pi\varepsilon_0 L\,\frac{2.5}{0.5} = 5\pi\varepsilon_0 L.$$

So

$$\boxed{C_{\text{var}} = 5\pi\varepsilon_0 L \approx 15.708\,\varepsilon_0 L.}$$

The ratio is

$$\frac{C_{\text{var}}}{C_{\text{exact}}} \approx 1.01366.$$

So the variational value is about

$$1.37\%$$

above the exact result.

5. Case $b/a = 2$

Now

$$\beta = 2.$$

The exact capacitance is

$$C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln 2}.$$

Since

$$\ln 2 \approx 0.693147,$$

we get

$$\boxed{C_{\text{exact}} \approx 9.0647\,\varepsilon_0 L.}$$

The variational result is

$$C_{\text{var}} = \pi\varepsilon_0 L\,\frac{1+2}{2-1} = 3\pi\varepsilon_0 L.$$

Thus

$$\boxed{C_{\text{var}} = 3\pi\varepsilon_0 L \approx 9.42478\,\varepsilon_0 L.}$$

The ratio is

$$\frac{C_{\text{var}}}{C_{\text{exact}}} \approx 1.03972.$$

So the variational estimate is about

$$3.97\%$$

too high.

6. Case $b/a = 3$

Now

$$\beta = 3.$$

The exact value is

$$C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln 3}.$$

Since

$$\ln 3 \approx 1.098612,$$

we obtain

$$\boxed{C_{\text{exact}} \approx 5.7192\,\varepsilon_0 L.}$$

The variational estimate is

$$C_{\text{var}} = \pi\varepsilon_0 L\,\frac{1+3}{3-1} = 2\pi\varepsilon_0 L.$$

Hence

$$\boxed{C_{\text{var}} = 2\pi\varepsilon_0 L \approx 6.28319\,\varepsilon_0 L.}$$

The ratio is

$$\frac{C_{\text{var}}}{C_{\text{exact}}} \approx 1.09861.$$

So the variational result is about

$$9.86\%$$

too high.

7. Summary table

b/a	$C_{\text{exact}}/(\varepsilon_0 L)$	$C_{\text{var}}/(\varepsilon_0 L)$	$C_{\text{var}}/C_{\text{exact}}$
1.5	15.496	15.708	1.01366
2	9.0647	9.42478	1.03972
3	5.7192	6.28319	1.09861

8. Why does the agreement get worse as b/a increases?

The exact potential for a coaxial cylindrical capacitor is

$$\Phi_{\text{exact}}(\rho) = \frac{\ln(b/\rho)}{\ln(b/a)}.$$

This is a logarithmic function of ρ.

The trial function used here is

$$\Psi_1(\rho) = \frac{b-\rho}{b-a},$$

which is linear in ρ.

So the reason for the trend is clear: the trial function has the correct boundary values, but not the correct radial dependence.

When b/a is close to 1: If the gap

$$b - a$$

is small, then the region between the cylinders is thin, and over a thin interval a logarithmic function is very nearly linear. Therefore the trial function is a very good approximation to the exact potential, and the variational upper bound is close to the exact capacitance.

When b/a becomes larger: As the annular region becomes wider, the exact logarithmic dependence differs more and more from a straight line. The trial function then becomes less accurate, so the variational functional gives a looser upper bound. That is why the percentage error increases as

$$\frac{b}{a}$$

increases.

9. Physical interpretation

The electrostatic energy density is proportional to

$$|\mathbf{E}|^2 = |\nabla\Phi|^2.$$

The exact field in a coaxial capacitor is

$$E_{\text{exact}}(\rho) = \frac{\text{const.}}{\rho},$$

because the exact potential is logarithmic. The trial function instead gives a constant radial derivative:

$$\frac{\mathrm{d}\Psi_1}{\mathrm{d}\rho} = -\frac{1}{b-a},$$

so it predicts a field of constant magnitude throughout the annulus.

That is physically incorrect. In the real capacitor the field is stronger near the inner cylinder and weaker near the outer cylinder. By forcing the field to be constant, the trial function distorts the energy distribution and produces a value of the energy that is too large. Since the variational principle gives

$$C \le \varepsilon_0 \int |\nabla\Psi|^2 \, d\tau,$$

this overestimate of the energy leads to an overestimate of the capacitance.

Thus the final result is

$$\boxed{C_{\text{var}} = \pi\varepsilon_0 L \frac{a+b}{b-a},}$$

which is an upper bound to

$$\boxed{C_{\text{exact}} = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}.}$$

For the three requested ratios:

$$\boxed{\begin{aligned} b/a = 1.5: &\quad C_{\text{var}} \approx 1.01366\, C_{\text{exact}}, \\ b/a = 2: &\quad C_{\text{var}} \approx 1.03972\, C_{\text{exact}}, \\ b/a = 3: &\quad C_{\text{var}} \approx 1.09861\, C_{\text{exact}}. \end{aligned}}$$

So the naive variational estimate is best when the gap is narrow and becomes increasingly poor as the annulus widens, because the exact potential is logarithmic while the trial function is only linear.

Problem 1.7

13.1 Problem Statement

Two long, cylindrical conductors of radii a_1 and a_2 are parallel and separated by a distance d, which is large compared with either radius. Show that the capacitance per unit length is given approximately by

$$C = \pi\varepsilon_0 \left(\ln \frac{d}{a}\right)^{-1},$$

where a is the geometrical mean of the two radii.

Approximately what gauge wire (state diameter in millimeters) would be necessary to make a two-wire transmission line with a capacitance of

$$1.2 \times 10^{-11}\ \text{F/m}$$

if the separation of the wires was 0.5 cm? 1.5 cm? 5.0 cm?

13.2 Solution

We consider two very long parallel cylindrical conductors carrying equal and opposite charges per unit length,

$$+\lambda \qquad \text{and} \qquad -\lambda.$$

Because the wires are very long, the problem is effectively two-dimensional in the plane perpendicular to the wires.

The idea is to approximate each cylindrical conductor by a line charge placed along its axis. This is valid because the separation

$$d$$

is assumed large compared with the radii

$$a_1,\ a_2.$$

So each wire is small compared with the distance to the other wire, and the field produced by one wire over the cross section of the other changes only slightly across that cross section.

1. Potential of a line charge

For an infinite line charge of density λ, Gauss's law gives

$$E(r) = \frac{\lambda}{2\pi\varepsilon_0 r},$$

directed radially outward.

Since

$$E_r = -\frac{\mathrm{d}\Phi}{\mathrm{d}r},$$

we have

$$-\frac{\mathrm{d}\Phi}{\mathrm{d}r} = \frac{\lambda}{2\pi\varepsilon_0 r}.$$

Hence

$$\frac{\mathrm{d}\Phi}{\mathrm{d}r} = -\frac{\lambda}{2\pi\varepsilon_0}\frac{1}{r}.$$

Integrating,

$$\Phi(r) = -\frac{\lambda}{2\pi\varepsilon_0} \ln r + \text{constant}.$$

So the potential produced by a line charge $+\lambda$ at distance r_1 is

$$\Phi_1 = -\frac{\lambda}{2\pi\varepsilon_0} \ln r_1 + \text{constant},$$

while the potential produced by a line charge $-\lambda$ at distance r_2 is

$$\Phi_2 = +\frac{\lambda}{2\pi\varepsilon_0} \ln r_2 + \text{constant}.$$

Therefore the total potential at a point is

$$\Phi = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{r_2}{r_1} + \text{constant}.$$

Here

- r_1 is the distance to the axis of the positively charged wire,
- r_2 is the distance to the axis of the negatively charged wire.

2. Potential on the surface of wire 1

Let wire 1 have radius a_1 and wire 2 have radius a_2.

On the surface of wire 1, the distance to its own axis is approximately

$$r_1 = a_1.$$

Because $d \gg a_1, a_2$, the distance from a point on wire 1 to the axis of wire 2 is approximately just

$$r_2 \approx d.$$

Therefore the potential on wire 1 is approximately

$$\Phi_1 \approx \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{d}{a_1} + \text{constant}.$$

3. Potential on the surface of wire 2

Similarly, on the surface of wire 2,

$$r_2 = a_2, \qquad r_1 \approx d.$$

Hence the potential there is approximately

$$\Phi_2 \approx \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{a_2}{d} + \text{constant} = -\frac{\lambda}{2\pi\varepsilon_0} \ln \frac{d}{a_2} + \text{constant}.$$

4. Potential difference

The potential difference between the wires is therefore

$$V = \Phi_1 - \Phi_2.$$

Substituting the two expressions,

$$V = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{d}{a_1} - \left(-\frac{\lambda}{2\pi\varepsilon_0} \ln \frac{d}{a_2} \right).$$

So

$$V = \frac{\lambda}{2\pi\varepsilon_0} \left[\ln \frac{d}{a_1} + \ln \frac{d}{a_2} \right].$$

Use the logarithm rule

$$\ln x + \ln y = \ln(xy),$$

to obtain

$$V = \frac{\lambda}{2\pi\varepsilon_0} \ln\left(\frac{d}{a_1}\frac{d}{a_2}\right) = \frac{\lambda}{2\pi\varepsilon_0} \ln\left(\frac{d^2}{a_1 a_2}\right).$$

Now define

$$a = \sqrt{a_1 a_2},$$

the geometrical mean of the two radii. Then

$$a^2 = a_1 a_2,$$

so

$$\frac{d^2}{a_1 a_2} = \frac{d^2}{a^2}.$$

Therefore

$$V = \frac{\lambda}{2\pi\varepsilon_0} \ln\left(\frac{d^2}{a^2}\right) = \frac{\lambda}{2\pi\varepsilon_0} \cdot 2\ln\frac{d}{a}.$$

Hence

$$\boxed{V = \frac{\lambda}{\pi\varepsilon_0} \ln\frac{d}{a}.}$$

5. Capacitance per unit length

The capacitance per unit length is

$$C = \frac{\lambda}{V}.$$

Substituting the expression for V,

$$C = \frac{\lambda}{\frac{\lambda}{\pi\varepsilon_0}\ln(d/a)}.$$

Cancel λ:

$$\boxed{C = \pi\varepsilon_0 \left(\ln\frac{d}{a}\right)^{-1}.}$$

This is the required result.

6. Special case of equal wires

If the two wires have the same radius,

$$a_1 = a_2 = r,$$

then their geometrical mean is just

$$a = \sqrt{r^2} = r.$$

So for equal wires,

$$\boxed{C = \pi\varepsilon_0 \left(\ln\frac{d}{r}\right)^{-1}.}$$

Since the problem asks what wire diameter is required, we now assume the two wires are identical, so

$$a = r = \text{wire radius}.$$

7. Solve for the wire radius

We are given

$$C = 1.2 \times 10^{-11}\ \text{F/m}.$$

From

$$C = \frac{\pi\varepsilon_0}{\ln(d/r)},$$

we get

$$\ln\frac{d}{r} = \frac{\pi\varepsilon_0}{C}.$$

Using

$$\varepsilon_0 = 8.854 \times 10^{-12}\ \text{F/m},$$

we have

$$\pi\varepsilon_0 = \pi(8.854 \times 10^{-12}) \approx 2.781 \times 10^{-11} \text{ F/m}.$$

Therefore

$$\ln \frac{d}{r} = \frac{2.781 \times 10^{-11}}{1.2 \times 10^{-11}} \approx 2.3175.$$

Exponentiating,

$$\frac{d}{r} = e^{2.3175} \approx 10.15.$$

So

$$r \approx \frac{d}{10.15}.$$

The wire diameter is

$$D = 2r \approx \frac{2d}{10.15} \approx 0.197\, d.$$

Thus the diameter is about 19.7% of the wire spacing.

8. Numerical values

(i) Separation $d = 0.5$ cm Convert to meters and millimeters:

$$d = 0.5 \text{ cm} = 5 \text{ mm}.$$

Then

$$D \approx 0.197(5 \text{ mm}) \approx 0.985 \text{ mm}.$$

So

$$\boxed{D \approx 0.99 \text{ mm} \approx 1.0 \text{ mm}.}$$

(ii) Separation $d = 1.5$ cm Now

$$d = 1.5 \text{ cm} = 15 \text{ mm}.$$

Hence

$$D \approx 0.197(15 \text{ mm}) \approx 2.96 \text{ mm}.$$

So

$$\boxed{D \approx 3.0 \text{ mm}.}$$

(iii) Separation $d = 5.0$ cm Now

$$d = 5.0 \text{ cm} = 50 \text{ mm}.$$

Then

$$D \approx 0.197(50 \text{ mm}) \approx 9.85 \text{ mm}.$$

So

$$\boxed{D \approx 9.9 \text{ mm}.}$$

9. Physical interpretation

The capacitance per unit length becomes larger when the wires are thicker, because thicker conductors allow more charge to be stored for the same potential difference. It becomes smaller when the wires are farther apart, because a larger separation requires a larger potential difference for the same line charge.

The logarithm appears because the electric field of a line charge varies as

$$E \propto \frac{1}{r},$$

so the potential difference between cylindrical surfaces involves the integral

$$\int \frac{dr}{r} = \ln r.$$

Final answer

$$\boxed{C = \pi\varepsilon_0 \left(\ln \frac{d}{a}\right)^{-1}, \qquad a = \sqrt{a_1 a_2}.}$$

For equal wires with

$$C = 1.2 \times 10^{-11}\ \mathrm{F/m},$$

the required diameters are approximately

$$\boxed{D \approx 1.0\ \mathrm{mm} \quad \text{for } d = 0.5\ \mathrm{cm},}$$

$$\boxed{D \approx 3.0\ \mathrm{mm} \quad \text{for } d = 1.5\ \mathrm{cm},}$$

and

$$\boxed{D \approx 9.9\ \mathrm{mm} \quad \text{for } d = 5.0\ \mathrm{cm}.}$$

Problem 1.18

14.1 Problem Statement

Consider the configuration of conductors of Problem 1.17, with all conductors except S_1 held at zero potential.

(a) Show that the potential $\Phi(\mathbf{x})$ anywhere in the volume V and on any of the surfaces S_i can be written

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}')\, G(\mathbf{x}, \mathbf{x}')\, da',$$

where $\sigma_1(\mathbf{x}')$ is the surface charge density on S_1 and $G(\mathbf{x}, \mathbf{x}')$ is the Green function potential for a point charge in the presence of all the surfaces that are held at zero potential (but with S_1 absent). Show also that the electrostatic energy is

$$W = \frac{1}{8\pi\varepsilon_0} \oint_{S_1} da \oint_{S_1} da'\, \sigma_1(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}'),$$

where the integrals are only over the surface S_1.

(b) Show that the variational expression

$$C^{-1}[\sigma] = \frac{\displaystyle\oint_{S_1} da \oint_{S_1} da'\, \sigma(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma(\mathbf{x}')}{4\pi\varepsilon_0 \left[\displaystyle\oint_{S_1} \sigma(\mathbf{x})\, da\right]^2}$$

with an arbitrary integrable function $\sigma(\mathbf{x})$ defined on S_1, is stationary for small variations of σ away from σ_1. Use Thomson's theorem to prove that the reciprocal of $C^{-1}[\sigma]$ gives a lower bound to the true capacitance of the conductor S_1.

14.2 Solution

We consider a set of conductors with surfaces

$$S_1, S_2, \ldots, S_n.$$

All conductors except S_1 are grounded:

$$\Phi = 0 \qquad \text{on } S_2, \ldots, S_n.$$

The conductor S_1 is held at some constant potential V_1 and carries surface charge density $\sigma_1(\mathbf{x})$.

The central idea is that, once the grounded conductors are fixed, the electrostatic field produced by the charge on S_1 can be described through the Green function of the region with all grounded surfaces present and S_1 removed.

Part (a): Representation of the potential

Step 1: Meaning of the Green function. Let

$$G(\mathbf{x}, \mathbf{x}')$$

be the Green function for the region when all grounded conductors are present but S_1 is absent. By definition, for fixed source point $\mathbf{x}'$ it satisfies

$$\nabla_{\mathbf{x}}^2 G(\mathbf{x}, \mathbf{x}') = -4\pi\delta(\mathbf{x} - \mathbf{x}'),$$

and

$$G(\mathbf{x}, \mathbf{x}') = 0 \qquad \text{on all grounded surfaces } S_2, \ldots, S_n.$$

Physically, $G(\mathbf{x}, \mathbf{x}')$ is the potential produced at $\mathbf{x}$ by a unit point charge placed at $\mathbf{x}'$ in the presence of those grounded conductors.

Therefore, if an infinitesimal surface element da' on S_1 carries charge

$$dq = \sigma_1(\mathbf{x}')\, da',$$

its contribution to the potential at $\mathbf{x}$ is

$$d\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} G(\mathbf{x}, \mathbf{x}')\, dq = \frac{1}{4\pi\varepsilon_0} G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}')\, da'.$$

By superposition, summing the contributions of all surface elements on S_1 gives

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}')\, G(\mathbf{x}, \mathbf{x}')\, da'.$$

Hence

$$\boxed{\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}')\, G(\mathbf{x}, \mathbf{x}')\, da'.}$$

This holds everywhere in the field region V, and by continuity of the potential it also holds on the conductor surfaces.

Step 2: Energy in terms of the surface charge density. The electrostatic energy can be written as

$$W = \frac{1}{2} \oint_{S_1} \sigma_1(\mathbf{x})\, \Phi(\mathbf{x})\, da,$$

because the only conductor at nonzero potential is S_1.

Now substitute the representation for $\Phi(\mathbf{x})$:

$$W = \frac{1}{2} \oint_{S_1} \sigma_1(\mathbf{x}) \left[\frac{1}{4\pi\varepsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}')\, G(\mathbf{x}, \mathbf{x}')\, da' \right] da.$$

Pull out the constant:

$$W = \frac{1}{8\pi\varepsilon_0} \oint_{S_1} da \oint_{S_1} da'\, \sigma_1(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}').$$

Therefore

$$\boxed{W = \frac{1}{8\pi\varepsilon_0} \oint_{S_1} da \oint_{S_1} da'\, \sigma_1(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}').}$$

That proves part **(a)**.

Part (b): Stationarity of $C^{-1}[\sigma]$

We now introduce the trial functional

$$C^{-1}[\sigma] = \frac{N[\sigma]}{4\pi\varepsilon_0\, Q[\sigma]^2},$$

where

$$N[\sigma] = \oint_{S_1} da \oint_{S_1} da'\, \sigma(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma(\mathbf{x}'),$$

and

$$Q[\sigma] = \oint_{S_1} \sigma(\mathbf{x})\, da.$$

We must show two things:

1. $C^{-1}[\sigma]$ is stationary when $\sigma = \sigma_1$ is the true charge density;
2. its reciprocal gives a lower bound to the true capacitance.

Step 1: Evaluate the functional at the true charge density. For the true surface charge density σ_1, from part **(a)** we have

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \oint_{S_1} G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}')\, da'.$$

But on the conductor S_1 itself, the potential is constant:

$$\Phi(\mathbf{x}) = V_1 \qquad \text{for } \mathbf{x} \in S_1.$$

Hence

$$\oint_{S_1} G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}')\, da' = 4\pi\varepsilon_0 V_1 \qquad (\mathbf{x} \in S_1).$$

Now compute

$$N[\sigma_1] = \oint_{S_1} da\, \sigma_1(\mathbf{x}) \oint_{S_1} da'\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}').$$

Using the previous relation,

$$N[\sigma_1] = \oint_{S_1} \sigma_1(\mathbf{x})\, (4\pi\varepsilon_0 V_1)\, da = 4\pi\varepsilon_0 V_1 Q_1,$$

where

$$Q_1 = \oint_{S_1} \sigma_1(\mathbf{x})\, da$$

is the total charge on the conductor.

Therefore

$$C^{-1}[\sigma_1] = \frac{4\pi\varepsilon_0 V_1 Q_1}{4\pi\varepsilon_0 Q_1^2} = \frac{V_1}{Q_1} = \frac{1}{C},$$

since

$$C = \frac{Q_1}{V_1}.$$

So the functional indeed reproduces the true inverse capacitance when evaluated at the true charge density:

$$\boxed{C^{-1}[\sigma_1] = \frac{1}{C}.}$$

Step 2: Show stationarity. Let

$$\sigma = \sigma_1 + \delta\sigma,$$

where $\delta\sigma$ is an arbitrary small variation.

We compute the first variation of

$$F[\sigma] \equiv C^{-1}[\sigma] = \frac{N[\sigma]}{4\pi\varepsilon_0 Q[\sigma]^2}.$$

First vary $Q[\sigma]$:

$$\delta Q = \oint_{S_1} \delta\sigma(\mathbf{x})\, da.$$

Next vary $N[\sigma]$. Since $G(\mathbf{x}, \mathbf{x}')$ is symmetric,

$$G(\mathbf{x}, \mathbf{x}') = G(\mathbf{x}', \mathbf{x}),$$

we have

$$\delta N = 2 \oint_{S_1} da\, \delta\sigma(\mathbf{x}) \oint_{S_1} da'\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}').$$

But for the true charge density,

$$\oint_{S_1} da'\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}') = 4\pi\varepsilon_0 V_1.$$

Hence

$$\delta N = 2 \oint_{S_1} \delta\sigma(\mathbf{x})\, (4\pi\varepsilon_0 V_1)\, da = 8\pi\varepsilon_0 V_1\, \delta Q.$$

Now vary the quotient:

$$\delta F = \frac{\delta N}{4\pi\varepsilon_0 Q_1^2} - \frac{N[\sigma_1]}{4\pi\varepsilon_0} \frac{2\delta Q}{Q_1^3}.$$

Substitute

$$\delta N = 8\pi\varepsilon_0 V_1\,\delta Q, \qquad N[\sigma_1] = 4\pi\varepsilon_0 V_1 Q_1.$$

Then

$$\delta F = \frac{8\pi\varepsilon_0 V_1\,\delta Q}{4\pi\varepsilon_0 Q_1^2} - \frac{4\pi\varepsilon_0 V_1 Q_1}{4\pi\varepsilon_0}\frac{2\delta Q}{Q_1^3}.$$

Simplify each term:

$$\delta F = \frac{2V_1\,\delta Q}{Q_1^2} - \frac{2V_1\,\delta Q}{Q_1^2} = 0.$$

Therefore

$$\boxed{\delta C^{-1}[\sigma] = 0 \qquad \text{at } \sigma = \sigma_1,}$$

so the functional is stationary for small variations away from the true surface charge density.

Lower bound from Thomson's theorem

We now prove that the reciprocal of this functional gives a lower bound to the true capacitance.

Step 1: Thomson's theorem. Thomson's theorem states that among all charge distributions on the conductors with the same total charge on each conductor, the true electrostatic distribution minimizes the electrostatic energy.

In the present problem only the charge on S_1 is nonzero. So among all trial surface densities $\sigma(\mathbf{x})$ on S_1 with fixed total charge

$$Q = \oint_{S_1} \sigma(\mathbf{x})\,da,$$

the true distribution σ_1 minimizes the energy.

From part **(a)**, the energy corresponding to a trial density σ is

$$W[\sigma] = \frac{1}{8\pi\varepsilon_0}\oint_{S_1} da \oint_{S_1} da'\,\sigma(\mathbf{x})\,G(\mathbf{x},\mathbf{x}')\,\sigma(\mathbf{x}').$$

That is,

$$W[\sigma] = \frac{1}{8\pi\varepsilon_0}N[\sigma].$$

For the true distribution,

$$W_{\text{true}} = \frac{Q^2}{2C}.$$

Thomson's theorem therefore implies

$$W[\sigma] \geq W_{\text{true}} \qquad \text{for fixed } Q.$$

So

$$\frac{1}{8\pi\varepsilon_0}N[\sigma] \geq \frac{Q^2}{2C}.$$

Multiply both sides by 2:

$$\frac{1}{4\pi\varepsilon_0}N[\sigma] \geq \frac{Q^2}{C}.$$

Now divide by Q^2:

$$\frac{N[\sigma]}{4\pi\varepsilon_0 Q^2} \geq \frac{1}{C}.$$

But the left-hand side is exactly $C^{-1}[\sigma]$. Hence

$$\boxed{C^{-1}[\sigma] \geq \frac{1}{C}.}$$

Taking reciprocals,

$$\boxed{\frac{1}{C^{-1}[\sigma]} \leq C.}$$

Thus the reciprocal of the variational expression gives a *lower bound* to the true capacitance:

$$\boxed{C_{\text{trial}} \equiv \frac{1}{C^{-1}[\sigma]} \leq C.}$$

Physical interpretation

The functional $C^{-1}[\sigma]$ is constructed from the Green-function interaction energy of a trial surface charge distribution on S_1. If the trial charge is not the true one, then for the same total charge it stores too much electrostatic energy, because the true charge distribution is the one that arranges itself to minimize the energy.

Since for fixed charge

$$W = \frac{Q^2}{2C},$$

a larger energy corresponds to a smaller capacitance. That is why the capacitance inferred from a trial charge distribution is always below the exact capacitance.

Final results

$$\boxed{\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}')\, G(\mathbf{x}, \mathbf{x}')\, da'}$$

$$\boxed{W = \frac{1}{8\pi\varepsilon_0} \oint_{S_1} da \oint_{S_1} da'\, \sigma_1(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma_1(\mathbf{x}')}$$

$$\boxed{C^{-1}[\sigma] = \frac{\displaystyle\oint_{S_1} da \oint_{S_1} da'\, \sigma(\mathbf{x})\, G(\mathbf{x}, \mathbf{x}')\, \sigma(\mathbf{x}')}{4\pi\varepsilon_0 \left[\displaystyle\oint_{S_1} \sigma(\mathbf{x})\, da\right]^2}}$$

is stationary at the true charge density σ_1, and

$$\boxed{C^{-1}[\sigma] \geq \frac{1}{C} \qquad \Longrightarrow \qquad \frac{1}{C^{-1}[\sigma]} \leq C.}$$

So the reciprocal of the variational expression gives a lower bound to the true capacitance.

Problem 1.8

15.1 Problem Statement

(a) For the three capacitor geometries in Problem 1.6 calculate the total electro- static energy and express it alternatively in terms of the equal and opposite charges Q and $-Q$ placed on the conductors and the potential difference between them.

(b) Sketch the energy density of the electrostatic field in each case as a function of the appropriate linear coordinate.

15.2 Solution

For any capacitor, the total electrostatic energy can be written in either of the equivalent forms

$$W = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV,$$

where

$$Q = \text{magnitude of the charge on either conductor}, \qquad V = \text{potential difference}.$$

It can also be written as a field energy:

$$W = \frac{\varepsilon_0}{2} \int E^2\, d\tau.$$

The corresponding energy density is

$$u = \frac{\varepsilon_0}{2} E^2.$$

We now apply this to the three geometries of Problem 1.6.

(a) Parallel-plate capacitor

Two large conducting plates of area A are separated by distance d, with charges $+Q$ and $-Q$.

From Problem 1.6,

$$C = \frac{\varepsilon_0 A}{d}, \qquad E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}, \qquad V = Ed = \frac{Qd}{\varepsilon_0 A}.$$

Energy from C and V. Using

$$W = \frac{1}{2}CV^2,$$

we get

$$W = \frac{1}{2}\left(\frac{\varepsilon_0 A}{d}\right)V^2.$$

Hence

$$\boxed{W = \frac{1}{2}\frac{\varepsilon_0 A}{d}V^2.}$$

Energy from Q and C. Using

$$W = \frac{Q^2}{2C},$$

we obtain

$$W = \frac{Q^2}{2(\varepsilon_0 A/d)} = \frac{Q^2 d}{2\varepsilon_0 A}.$$

Thus

$$\boxed{W = \frac{Q^2 d}{2\varepsilon_0 A}.}$$

Energy from the field integral. Since the field is essentially uniform between the plates and negligible outside,

$$u = \frac{\varepsilon_0}{2}E^2 = \frac{\varepsilon_0}{2}\left(\frac{Q}{\varepsilon_0 A}\right)^2 = \frac{Q^2}{2\varepsilon_0 A^2}.$$

The field occupies the volume

$$Ad.$$

Therefore

$$W = u(Ad) = \frac{Q^2}{2\varepsilon_0 A^2}(Ad) = \frac{Q^2 d}{2\varepsilon_0 A},$$

in agreement with the previous result.

So for the parallel-plate capacitor

$$\boxed{W = \frac{1}{2}\frac{\varepsilon_0 A}{d}V^2 = \frac{Q^2 d}{2\varepsilon_0 A} = \frac{1}{2}QV.}$$

Energy density. Since the field is constant between the plates,

$$u = \frac{\varepsilon_0}{2}E^2 = \text{constant} \qquad (0 < x < d),$$

if x is the coordinate normal to the plates, and

$$u \approx 0 \qquad \text{outside.}$$

Therefore the energy density as a function of x is approximately a rectangular step: constant between the plates, nearly zero outside.

(b) Spherical capacitor

Now consider two concentric conducting spheres of radii a and b with $b > a$.

From Problem 1.6,

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a},$$

and in the region

$$a < r < b$$

the electric field is

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}.$$

The potential difference is

$$V = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right).$$

Energy from C and V. Using

$$W = \frac{1}{2}CV^2,$$

we get

$$W = \frac{1}{2}\left(4\pi\varepsilon_0 \frac{ab}{b-a}\right)V^2.$$

So

$$\boxed{W = 2\pi\varepsilon_0 \frac{ab}{b-a}V^2.}$$

Energy from Q and C. Using

$$W = \frac{Q^2}{2C},$$

we find

$$W = \frac{Q^2}{2 \cdot 4\pi\varepsilon_0\, ab/(b-a)} = \frac{Q^2(b-a)}{8\pi\varepsilon_0 ab}.$$

Therefore

$$\boxed{W = \frac{Q^2(b-a)}{8\pi\varepsilon_0 ab}.}$$

Energy from the field integral. The energy density is

$$u(r) = \frac{\varepsilon_0}{2}E(r)^2 = \frac{\varepsilon_0}{2}\left(\frac{Q}{4\pi\varepsilon_0 r^2}\right)^2 = \frac{Q^2}{32\pi^2\varepsilon_0 r^4}.$$

The volume element in spherical coordinates is

$$d\tau = 4\pi r^2\, dr.$$

Hence

$$W = \int_a^b u(r)\, 4\pi r^2\, dr = \int_a^b \frac{Q^2}{32\pi^2\varepsilon_0 r^4}\, 4\pi r^2\, dr.$$

Simplify:

$$W = \frac{Q^2}{8\pi\varepsilon_0}\int_a^b \frac{dr}{r^2}.$$

Since

$$\int_a^b \frac{dr}{r^2} = \left[-\frac{1}{r}\right]_a^b = \frac{1}{a} - \frac{1}{b},$$

we obtain

$$W = \frac{Q^2}{8\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right) = \frac{Q^2(b-a)}{8\pi\varepsilon_0 ab},$$

which agrees with the previous result.

Thus for the spherical capacitor

$$\boxed{W = 2\pi\varepsilon_0 \frac{ab}{b-a}V^2 = \frac{Q^2(b-a)}{8\pi\varepsilon_0 ab} = \frac{1}{2}QV.}$$

Energy density. The energy density is

$$\boxed{u(r) = \frac{Q^2}{32\pi^2\varepsilon_0 r^4}, \qquad a < r < b.}$$

So it decreases very rapidly with radius as

$$u(r) \propto \frac{1}{r^4}.$$

It is largest near the inner sphere and smallest near the outer sphere.

(c) Cylindrical capacitor

Now consider two concentric conducting cylinders of radii a and b, length L, with

$$b > a.$$

From Problem 1.6,

$$C = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}.$$

The electric field in the region

$$a < r < b$$

is

$$E(r) = \frac{Q}{2\pi\varepsilon_0 L}\frac{1}{r}.$$

The potential difference is

$$V = \frac{Q}{2\pi\varepsilon_0 L}\ln\frac{b}{a}.$$

Energy from C and V. Using

$$W = \frac{1}{2}CV^2,$$

we find

$$W = \frac{1}{2}\left(\frac{2\pi\varepsilon_0 L}{\ln(b/a)}\right)V^2.$$

Therefore

$$\boxed{W = \frac{\pi\varepsilon_0 L}{\ln(b/a)}V^2.}$$

Energy from Q and C. Using

$$W = \frac{Q^2}{2C},$$

we obtain

$$W = \frac{Q^2}{2\cdot 2\pi\varepsilon_0 L/\ln(b/a)} = \frac{Q^2}{4\pi\varepsilon_0 L}\ln\frac{b}{a}.$$

So

$$\boxed{W = \frac{Q^2}{4\pi\varepsilon_0 L}\ln\frac{b}{a}.}$$

Energy from the field integral. The energy density is

$$u(r) = \frac{\varepsilon_0}{2}E(r)^2 = \frac{\varepsilon_0}{2}\left(\frac{Q}{2\pi\varepsilon_0 L r}\right)^2 = \frac{Q^2}{8\pi^2\varepsilon_0 L^2 r^2}.$$

The volume element for a cylindrical shell of radius r and thickness dr is

$$d\tau = 2\pi r L\,dr.$$

Hence

$$W = \int_a^b u(r)\,2\pi r L\,dr.$$

Substitute $u(r)$:

$$W = \int_a^b \frac{Q^2}{8\pi^2\varepsilon_0 L^2 r^2}(2\pi r L)\,dr.$$

Simplify:

$$W = \frac{Q^2}{4\pi\varepsilon_0 L}\int_a^b \frac{dr}{r}.$$

Since

$$\int_a^b \frac{dr}{r} = \ln\frac{b}{a},$$

we obtain

$$W = \frac{Q^2}{4\pi\varepsilon_0 L}\ln\frac{b}{a},$$

again in agreement.

Thus for the cylindrical capacitor

$$\boxed{W = \frac{\pi\varepsilon_0 L}{\ln(b/a)}V^2 = \frac{Q^2}{4\pi\varepsilon_0 L}\ln\frac{b}{a} = \frac{1}{2}QV.}$$

Energy density. The energy density is

$$\boxed{u(r) = \frac{Q^2}{8\pi^2\varepsilon_0 L^2 r^2}, \qquad a < r < b.}$$

So it decreases with radius as

$$u(r) \propto \frac{1}{r^2}.$$

It is largest near the inner cylinder and decreases more slowly than in the spherical case.

Part (b): Sketches of u

The appropriate linear coordinate in each case is:

- parallel plates: the perpendicular distance x between plates,
- spherical capacitor: the radial coordinate r,
- cylindrical capacitor: the radial coordinate r.

Therefore the qualitative behaviors are:

Parallel plates:

$$u(x) = \text{constant} \qquad (0 < x < d),$$

and essentially zero outside. The sketch is a flat horizontal line between the plates.

Spherical capacitor:

$$u(r) \propto \frac{1}{r^4}, \qquad a < r < b.$$

The sketch starts at a large value at $r = a$, then falls off steeply as r increases.

Cylindrical capacitor:

$$u(r) \propto \frac{1}{r^2}, \qquad a < r < b.$$

The sketch also decreases with r, but much more slowly than the spherical case.

Final answers

Parallel plates

$$\boxed{W = \frac{1}{2}\frac{\varepsilon_0 A}{d}V^2 = \frac{Q^2 d}{2\varepsilon_0 A} = \frac{1}{2}QV}$$

with

$$\boxed{u = \frac{\varepsilon_0}{2}E^2 = \frac{Q^2}{2\varepsilon_0 A^2} \quad \text{(constant between plates).}}$$

Spherical capacitor

$$\boxed{W = 2\pi\varepsilon_0 \frac{ab}{b-a}V^2 = \frac{Q^2(b-a)}{8\pi\varepsilon_0 ab} = \frac{1}{2}QV}$$

with

$$\boxed{u(r) = \frac{Q^2}{32\pi^2\varepsilon_0 r^4}.}$$

Cylindrical capacitor

$$\boxed{W = \frac{\pi\varepsilon_0 L}{\ln(b/a)}V^2 = \frac{Q^2}{4\pi\varepsilon_0 L}\ln\frac{b}{a} = \frac{1}{2}QV}$$

with

$$\boxed{u(r) = \frac{Q^2}{8\pi^2\varepsilon_0 L^2 r^2}.}$$

Problem 1.17

16.1 Problem Statement

A volume V in vacuum is bounded by a surface S consisting of several separate conducting surfaces S_i. One conductor is held at unit potential and all the other conductors at zero potential.

(a) Show that the capacitance of the one conductor is given by

$$C = \varepsilon_0 \int_V |\nabla\Phi|^2 \, d^3x$$

where $\Phi(\mathbf{x})$ is the solution for the potential.

(b) Show that the true capacitance C is always less than or equal to the quantity

$$C[\Psi] = \varepsilon_0 \int_V |\nabla\Psi|^2 \, d^3x$$

where Ψ is any trial function satisfying the boundary conditions on the conductors. This is a variational principle for the capacitance that yields an upper bound.

16.2 Solution

We consider a region V of vacuum bounded by conducting surfaces

$$S = S_1 \cup S_2 \cup \cdots,$$

with boundary conditions

$$\Phi = 1 \quad \text{on one conductor, say } S_1, \qquad \Phi = 0 \quad \text{on all other conductors.}$$

Since there is no free charge in the vacuum region between the conductors, the exact potential satisfies Laplace's equation:

$$\nabla^2\Phi = 0 \qquad \text{in } V.$$

The capacitance is defined by

$$C = \frac{Q}{V_0}.$$

But here the conductor S_1 is held at *unit* potential and the others at zero, so the potential difference is

$$V_0 = 1.$$

Therefore

$$\boxed{C = Q,}$$

where Q is the total charge on the conductor at unit

Problem 1.9

17.1 Problem Statement

Calculate the attractive force between conductors in the parallel plate capacitor (Problem 1.6a) and the parallel cylinder capacitor (Problem 1.7) for

(a) fixed charges on each conductor;

(b) fixed potential difference between conductors.

17.2 Solution

The force can be obtained from the electrostatic energy. If the geometry depends on a single separation parameter d, then the mechanical force in the direction of increasing d is

$$F_d = -\frac{\mathrm{d}W}{\mathrm{d}d}.$$

If this derivative is negative, the force tends to decrease d, which means the force is attractive. In what follows, we will compute the force and then state its magnitude.

There is one important distinction:

- For **fixed charge**, the relevant energy is the electrostatic energy stored in the field,

$$W = \frac{Q^2}{2C}.$$

- For **fixed potential difference**, the conductors are connected to a battery, so charge can flow. Then the correct quantity for the mechanical force is

$$W_{\text{eff}} = W - VQ = \frac{1}{2}CV^2 - CV^2 = -\frac{1}{2}CV^2.$$

This second formula is essential. If one differentiates only the field energy

$$W = \frac{1}{2}CV^2$$

while holding V fixed, one neglects the work done by the battery.

(1) Parallel-plate capacitor

For two large parallel plates of area A separated by distance d, the capacitance is

$$C = \frac{\varepsilon_0 A}{d}.$$

(a) Fixed charges $\pm Q$ For fixed Q,

$$W = \frac{Q^2}{2C}.$$

Substitute

$$C = \frac{\varepsilon_0 A}{d}:$$

$$W = \frac{Q^2}{2(\varepsilon_0 A/d)} = \frac{Q^2 d}{2\varepsilon_0 A}.$$

Now differentiate with respect to d:

$$F_d = -\frac{\mathrm{d}W}{\mathrm{d}d} = -\frac{\mathrm{d}}{\mathrm{d}d}\left(\frac{Q^2 d}{2\varepsilon_0 A}\right) = -\frac{Q^2}{2\varepsilon_0 A}.$$

The negative sign means the force is toward smaller d, so it is attractive. Its magnitude is

$$\boxed{|F| = \frac{Q^2}{2\varepsilon_0 A}.}$$

It is often convenient to express this in terms of the surface charge density

$$\sigma = \frac{Q}{A}.$$

Then

$$|F| = \frac{\sigma^2 A}{2\varepsilon_0}.$$

So for fixed charge,

$$\boxed{F_{\text{plates}} = -\frac{Q^2}{2\varepsilon_0 A} \quad \text{(toward smaller } d\text{)}, \qquad |F| = \frac{Q^2}{2\varepsilon_0 A}.}$$

Physical check. The electric field produced by one plate at the location of the other is

$$E_{\text{other}} = \frac{\sigma}{2\varepsilon_0}.$$

Hence the force on one plate is

$$F = QE_{\text{other}} = (\sigma A)\frac{\sigma}{2\varepsilon_0} = \frac{\sigma^2 A}{2\varepsilon_0},$$

which agrees with the energy method.

(b) Fixed potential difference V Now the conductors are maintained at fixed potential difference V. The effective energy is

$$W_{\text{eff}} = -\frac{1}{2}CV^2.$$

Using

$$C = \frac{\varepsilon_0 A}{d},$$

we get

$$W_{\text{eff}} = -\frac{1}{2}\frac{\varepsilon_0 A}{d}V^2.$$

Differentiate:

$$F_d = -\frac{\mathrm{d}W_{\text{eff}}}{\mathrm{d}d} = -\frac{\mathrm{d}}{\mathrm{d}d}\left(-\frac{1}{2}\frac{\varepsilon_0 A}{d}V^2\right).$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}d}\left(\frac{1}{d}\right) = -\frac{1}{d^2},$$

we have

$$\frac{\mathrm{d}}{\mathrm{d}d}\left(-\frac{1}{2}\frac{\varepsilon_0 A}{d}V^2\right) = +\frac{1}{2}\frac{\varepsilon_0 A V^2}{d^2}.$$

Therefore

$$F_d = -\frac{1}{2}\frac{\varepsilon_0 A V^2}{d^2}.$$

Again the negative sign means the force is attractive. Its magnitude is

$$\boxed{|F| = \frac{\varepsilon_0 A V^2}{2d^2}.}$$

Since

$$E = \frac{V}{d},$$

this can also be written as

$$\boxed{|F| = \frac{\varepsilon_0}{2} E^2 A.}$$

Thus for fixed potential difference,

$$\boxed{F_{\text{plates}} = -\frac{\varepsilon_0 A V^2}{2d^2} \quad \text{(toward smaller } d\text{)}, \qquad |F| = \frac{\varepsilon_0 A V^2}{2d^2}.}$$

(2) Parallel cylindrical conductors

Now consider the two-wire system of Problem 1.7: two long parallel cylindrical conductors, separated by distance d, with

$$d \gg a_1,\ a_2.$$

The capacitance per unit length is approximately

$$C' = \pi\varepsilon_0 \left(\ln\frac{d}{a}\right)^{-1},$$

where

$$a = \sqrt{a_1 a_2}.$$

Because the conductors are long, the total capacitance and total force are both proportional to length. It is therefore most natural to compute the **force per unit length**.

Let the charge per unit length on the two wires be

$$+\lambda, \qquad -\lambda.$$

(a) Fixed charges $\pm\lambda$ per unit length The electrostatic energy per unit length is

$$w = \frac{\lambda^2}{2C'}.$$

Substitute

$$C' = \frac{\pi\varepsilon_0}{\ln(d/a)}:$$

$$w = \frac{\lambda^2}{2}\frac{\ln(d/a)}{\pi\varepsilon_0} = \frac{\lambda^2}{2\pi\varepsilon_0}\ln\frac{d}{a}.$$

Now differentiate with respect to d:

$$f_d = -\frac{\mathrm{d}w}{\mathrm{d}d} = -\frac{\lambda^2}{2\pi\varepsilon_0}\frac{\mathrm{d}}{\mathrm{d}d}\ln\frac{d}{a}.$$

Since a is constant,

$$\frac{\mathrm{d}}{\mathrm{d}d}\ln\frac{d}{a} = \frac{1}{d}.$$

Hence

$$f_d = -\frac{\lambda^2}{2\pi\varepsilon_0 d}.$$

So the force per unit length is attractive, with magnitude

$$\boxed{\frac{|F|}{L} = \frac{\lambda^2}{2\pi\varepsilon_0 d}.}$$

Thus

$$\boxed{\frac{F_{\text{cyl}}}{L} = -\frac{\lambda^2}{2\pi\varepsilon_0 d} \quad \text{(toward smaller } d\text{)}, \qquad \frac{|F|}{L} = \frac{\lambda^2}{2\pi\varepsilon_0 d}.}$$

Physical check. The field at one wire due to the other wire is approximately that of a line charge:

$$E = \frac{\lambda}{2\pi\varepsilon_0 d}.$$

So the force per unit length is

$$\frac{F}{L} = \lambda E = \frac{\lambda^2}{2\pi\varepsilon_0 d},$$

again in agreement.

(b) Fixed potential difference V Now the wires are kept at fixed potential difference V. The effective energy per unit length is

$$w_{\text{eff}} = -\frac{1}{2}C'V^2.$$

Substitute

$$C' = \pi\varepsilon_0 \left(\ln\frac{d}{a}\right)^{-1},$$

to obtain

$$w_{\text{eff}} = -\frac{1}{2}\pi\varepsilon_0 V^2 \left(\ln\frac{d}{a}\right)^{-1}.$$

Differentiate:

$$f_d = -\frac{\mathrm{d}w_{\text{eff}}}{\mathrm{d}d} = -\frac{\mathrm{d}}{\mathrm{d}d}\left[-\frac{1}{2}\pi\varepsilon_0 V^2 \left(\ln\frac{d}{a}\right)^{-1}\right].$$

Now

$$\frac{\mathrm{d}}{\mathrm{d}d}\left(\ln\frac{d}{a}\right)^{-1} = -\left(\ln\frac{d}{a}\right)^{-2}\frac{\mathrm{d}}{\mathrm{d}d}\ln\frac{d}{a} = -\frac{1}{d\,\ln^2(d/a)}.$$

Therefore

$$\frac{\mathrm{d}w_{\text{eff}}}{\mathrm{d}d} = -\frac{1}{2}\pi\varepsilon_0 V^2\left(-\frac{1}{d\,\ln^2(d/a)}\right) = \frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)}.$$

Hence

$$f_d = -\frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)}.$$

So the force per unit length is again attractive, with magnitude

$$\boxed{\frac{|F|}{L} = \frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)}.}$$

Thus

$$\boxed{\frac{F_{\text{cyl}}}{L} = -\frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)} \quad \text{(toward smaller } d\text{)}, \qquad \frac{|F|}{L} = \frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)}.}$$

We can also write this in terms of the line charge density

$$\lambda = C'V = \frac{\pi\varepsilon_0 V}{\ln(d/a)}.$$

Then

$$\frac{|F|}{L} = \frac{\lambda^2}{2\pi\varepsilon_0 d},$$

which is the same formal expression as in the fixed-charge case, except that now λ itself changes when d changes because V is held fixed instead.

Final answers

Parallel-plate capacitor For fixed charges $\pm Q$:

$$\boxed{F = -\frac{Q^2}{2\varepsilon_0 A} \quad \text{(attractive)}, \qquad |F| = \frac{Q^2}{2\varepsilon_0 A}.}$$

For fixed potential difference V:

$$\boxed{F = -\frac{\varepsilon_0 A V^2}{2d^2} \quad \text{(attractive)}, \qquad |F| = \frac{\varepsilon_0 A V^2}{2d^2}.}$$

Parallel cylindrical conductors For fixed charges $\pm\lambda$ per unit length:

$$\boxed{\frac{F}{L} = -\frac{\lambda^2}{2\pi\varepsilon_0 d} \quad \text{(attractive)}, \qquad \frac{|F|}{L} = \frac{\lambda^2}{2\pi\varepsilon_0 d}.}$$

For fixed potential difference V:

$$\boxed{\frac{F}{L} = -\frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)} \quad \text{(attractive)}, \qquad \frac{|F|}{L} = \frac{\pi\varepsilon_0 V^2}{2d\,\ln^2(d/a)}.}$$

Problem 1.16

18.1 Problem Statement

Prove the following theorem: If a number of conducting surfaces are fixed in position with a given total charge on each, the introduction of an uncharged, insulated conductor into the region bounded by the surfaces lowers the electrostatic energy.

18.2 Solution

We are asked to prove an energy theorem in electrostatics.

The situation is this:

- A set of conductors is held fixed in space.
- Each conductor carries a specified total charge.
- The conductors produce some electrostatic field $\mathbf{E}$ in the surrounding region.
- Then we insert an additional conductor into the field region.
- This inserted conductor is *uncharged* and *insulated*, so its net charge remains zero.

We must show that after the conductor is inserted and electrostatic equilibrium is re-established, the total electrostatic energy is smaller than before.

The physical idea is that when the new conductor is inserted, charges are induced on its surface. These induced charges rearrange the field so that the field vanishes inside the new conductor. That removes some field-filled volume and therefore lowers the total field energy. We now prove this carefully.

1. Electrostatic energy in terms of the field

In vacuum, the electrostatic energy is

$$W = \frac{\varepsilon_0}{2}\int_{\text{all space}} E^2\,d\tau.$$

Equivalently, if Φ is the electrostatic potential,

$$\mathbf{E} = -\nabla\Phi,$$

so

$$W = \frac{\varepsilon_0}{2}\int |\nabla\Phi|^2\,d\tau.$$

We will compare two electrostatic configurations:

- the *original* configuration, before the new conductor is inserted;
- the *final* configuration, after insertion and equilibration.

Let

$$\Phi_0, \quad \mathbf{E}_0 = -\nabla\Phi_0, \quad W_0$$

denote the potential, field, and energy before insertion, and let

$$\Phi, \quad \mathbf{E} = -\nabla\Phi, \quad W$$

denote the corresponding quantities after insertion.

We must prove

$$\boxed{W \le W_0.}$$

2. The appropriate theorem to use

The key theorem is **Thomson's theorem**:

Among all charge distributions on a set of conductors having specified total charges on each conductor, the electrostatic equilibrium distribution is the one of minimum electrostatic energy.

So if we can construct *any* admissible trial charge distribution after the new conductor is inserted, and that trial distribution has the same total charge on each conductor as the original problem, then the true equilibrium energy after insertion must be less than or equal to the energy of that trial distribution.

That is exactly the strategy.

3. Construct a trial distribution after the conductor is inserted

Suppose the new conductor occupies a volume v, bounded by a surface s.

Before insertion, the original field $\mathbf{E}_0$ fills this region just as it fills the rest of space. Now imagine placing the new conductor into that region.

We now define a *trial* charge distribution for the new configuration as follows:

1. Keep all the original conductors with exactly the same total charges they had before.
2. Put zero net charge on the new conductor.
3. Arrange charges on the surface s of the new conductor in such a way that the potential inside the conductor becomes constant, so that the electric field inside it vanishes.

This is an admissible electrostatic configuration because:

- the original conductors still have the same given total charges,
- the inserted conductor is insulated and uncharged, so its total charge must be zero,
- the field inside a conductor in equilibrium must vanish.

By Thomson's theorem, the *actual* equilibrium after insertion has energy less than or equal to the energy of this trial configuration. So it is enough to show that this trial configuration already has energy not larger than W_0.

4. A convenient way to compare the energies

A clean proof comes from Green's first identity. For any sufficiently smooth scalar function Φ,

$$\int_V |\nabla\Phi|^2 \, d\tau = \oint_{\partial V} \Phi \, \frac{\partial \Phi}{\partial n} \, da - \int_V \Phi \, \nabla^2\Phi \, d\tau.$$

In electrostatics in vacuum, away from the conductor surfaces,

$$\nabla^2\Phi = 0.$$

So in the field region,

$$\int_V |\nabla\Phi|^2 \, d\tau = \oint_{\partial V} \Phi \, \frac{\partial \Phi}{\partial n} \, da.$$

But for a conductor surface,

$$\sigma = -\varepsilon_0 \, \frac{\partial \Phi}{\partial n},$$

where $\partial/\partial n$ is the derivative along the outward normal from the field region. Therefore

$$\varepsilon_0 \oint \Phi \, \frac{\partial \Phi}{\partial n} \, da = -\oint \Phi \, \sigma \, da.$$

Accounting consistently for normal directions gives the standard energy formula

$$W = \frac{1}{2} \sum_i Q_i V_i,$$

where Q_i is the total charge on conductor i, and V_i is its potential.

This formula is extremely useful here.

5. Energy before insertion

Before insertion, the conductors have charges $Q_1, Q_2, \ldots, Q_N$, fixed by the problem, and potentials $V_1^{(0)}, V_2^{(0)}, \ldots, V_N^{(0)}$. Therefore

$$W_0 = \frac{1}{2}\sum_{i=1}^{N} Q_i V_i^{(0)}.$$

6. Energy after insertion

After inserting the new conductor, the original conductors still have the same total charges $Q_1, \ldots, Q_N$. The new conductor has total charge

$$Q_{N+1} = 0.$$

Let the final conductor potentials be $V_1, V_2, \ldots, V_N$, and let the inserted conductor be at potential V_{N+1} (some constant, not specified in advance).

Then the electrostatic energy of the final equilibrium is

$$W = \frac{1}{2}\sum_{i=1}^{N+1} Q_i V_i.$$

Since the new conductor is uncharged,

$$Q_{N+1} = 0,$$

so

$$W = \frac{1}{2}\sum_{i=1}^{N} Q_i V_i.$$

Thus the only way the energy can change is through the changes in the potentials of the original conductors.

7. Why the energy must decrease

Now we invoke Thomson's theorem directly.

Take the original equilibrium field $\mathbf{E}_0$ and imagine removing from space the volume v now occupied by the inserted conductor. Inside that conductor the electrostatic field in the new equilibrium must be zero. So the field energy that originally occupied v,

$$\frac{\varepsilon_0}{2}\int_v E_0^2\, d\tau,$$

is eliminated.

Of course, inserting the conductor also changes the field everywhere outside v, because induced surface charges appear on the new conductor and slightly redistribute the fields in the rest of space. But the important point is this:

- the original field configuration $\mathbf{E}_0$ restricted to the exterior region is not the equilibrium configuration for the new set of conductors;
- the true equilibrium for the new geometry is the one that minimizes the total energy subject to the fixed conductor charges.

Therefore the new equilibrium energy cannot exceed the energy of any admissible post-insertion configuration, and in particular it cannot exceed what one would get by simply removing the old field energy from the conductor volume and then adjusting the surface charges consistently.

Hence the final energy must satisfy

$$W \leq W_0.$$

To see the inequality in a more explicitly variational way, let the inserted conductor occupy volume v. Then the final equilibrium field is the minimum-energy field among all fields consistent with:

- the same total charges on the original conductors,
- zero net charge on the inserted conductor,
- zero electric field inside the inserted conductor.

Because the condition $\mathbf{E} = 0$ in v removes a region where the original field had nonnegative energy density

$$u = \frac{\varepsilon_0}{2} E^2 \geq 0,$$

the minimizing field in the new class must have energy no greater than the old equilibrium field. So

$$\boxed{W \leq W_0.}$$

If the original field in the region where the conductor is inserted is not identically zero, then a strictly positive amount of field energy is removed from that region, and in that generic case

$$\boxed{W < W_0.}$$

Equality can occur only in the trivial case that the inserted conductor is placed entirely in a region where the original electric field already vanishes.

8. Alternative differential argument using polarization work

There is also a very intuitive physical derivation.

When the neutral insulated conductor is inserted, the external field induces surface charges on it: negative charge appears on the side facing the direction from which field lines enter, and positive charge appears on the opposite side. The conductor becomes polarized, though its net charge remains zero.

Because opposite induced charges are pulled in opposite directions by the electric field, the conductor is drawn into regions of stronger field. If we insert it quasistatically, the field does positive work on the conductor. By energy conservation, that work must come from the electrostatic field energy. Therefore the field energy decreases.

So physically:

$$\boxed{\text{the introduction of a neutral insulated conductor lowers the electrostatic energy.}}$$

9. Final statement

We conclude that if a system of conductors has fixed total charge on each conductor, then after inserting an additional uncharged insulated conductor and allowing the system to reach electrostatic equilibrium, the new energy satisfies

$$\boxed{W \leq W_0.}$$

In general the inequality is strict:

$$\boxed{W < W_0}$$

unless the conductor is inserted only in a region where the original electric field was zero.

Therefore the theorem is proved:

$$\boxed{\text{The introduction of an uncharged, insulated conductor lowers the electrostatic energy.}}$$

Problem 1.10

19.1 Problem Statement

Prove the mean value theorem: For charge-free space the value of the electrostatic potential at any point is equal to the average of the potential over the surface of any sphere centered on that point.

19.2 Solution

We must prove that if the electrostatic potential $\Phi(\mathbf{r})$ is harmonic in a region containing a sphere of radius R centered at a point $\mathbf{r}_0$, that is,

$$\nabla^2 \Phi = 0 \qquad \text{inside and on the sphere,}$$

then

$$\boxed{\Phi(\mathbf{r}_0) = \frac{1}{4\pi R^2} \oint_{|\mathbf{r}-\mathbf{r}_0|=R} \Phi(\mathbf{r})\, da.}$$

This is the mean value theorem for harmonic functions in electrostatics.

1. Physical setting

In charge-free space, Poisson's equation reduces to Laplace's equation:

$$\nabla^2 \Phi = 0.$$

So the electrostatic potential in any region containing no charge is a harmonic function.

The theorem states that at the center of any sphere lying entirely in such a charge-free region, the potential equals the average of the potential over the spherical surface.

This is a very characteristic property of harmonic functions. Physically it means that, in empty space, the potential at a point cannot be a strict local maximum or minimum produced by sources inside that region; it is determined in a balanced way by its surroundings.

2. Set up spherical coordinates about the point of interest

Take the point of interest as the origin of a local spherical coordinate system. That is, without loss of generality, let

$$\mathbf{r}_0 = 0.$$

Then we want to prove

$$\Phi(0) = \frac{1}{4\pi R^2} \oint_{r=R} \Phi \, da.$$

Since on a sphere of radius R the surface element is

$$da = R^2 \, d\Omega,$$

where

$$d\Omega = \sin\theta \, d\theta \, d\phi$$

is the solid-angle element, the statement is equivalently

$$\boxed{\Phi(0) = \frac{1}{4\pi} \int \Phi(R, \theta, \phi) \, d\Omega.}$$

So it is enough to show that the angular average of Φ on the sphere $r = R$ equals the value at the center.

3. General harmonic solution inside a sphere

Because Φ satisfies Laplace's equation in a region containing the sphere and its interior, the general solution regular at the origin can be expanded in spherical harmonics:

$$\Phi(r, \theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} A_{\ell m} \, r^\ell \, Y_{\ell m}(\theta, \phi).$$

We do *not* include the singular terms proportional to $r^{-(\ell+1)}$, because the potential is assumed finite at the center and harmonic throughout the sphere.

Now evaluate this at the center $r = 0$. Every term with $\ell \geq 1$ contains a factor r^ℓ, so those terms vanish at $r = 0$. Therefore

$$\Phi(0) = A_{00} Y_{00}.$$

Thus the value at the center is exactly the $\ell = 0$ part of the harmonic expansion.

4. Average the potential over the sphere

Now average $\Phi(R, \theta, \phi)$ over the sphere:

$$\frac{1}{4\pi} \int \Phi(R, \theta, \phi) \, d\Omega = \frac{1}{4\pi} \int \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} A_{\ell m} R^\ell Y_{\ell m}(\theta, \phi) \, d\Omega.$$

Interchange sum and integral:

$$\frac{1}{4\pi} \int \Phi(R, \theta, \phi) \, d\Omega = \frac{1}{4\pi} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} A_{\ell m} R^\ell \int Y_{\ell m}(\theta, \phi) \, d\Omega.$$

Now use the fact that all spherical harmonics except the constant one have zero average over the sphere:

$$\int Y_{\ell m}\, d\Omega = 0 \qquad \text{for } \ell \geq 1.$$

So only the $\ell = 0$, $m = 0$ term survives. Hence

$$\frac{1}{4\pi}\int \Phi(R,\theta,\phi)\, d\Omega = \frac{1}{4\pi} A_{00} Y_{00} \int d\Omega.$$

Since

$$\int d\Omega = 4\pi,$$

we get

$$\frac{1}{4\pi}\int \Phi(R,\theta,\phi)\, d\Omega = A_{00} Y_{00}.$$

But from above,

$$A_{00} Y_{00} = \Phi(0).$$

Therefore

$$\boxed{\Phi(0) = \frac{1}{4\pi}\int \Phi(R,\theta,\phi)\, d\Omega.}$$

Rewriting this in surface-integral form,

$$\boxed{\Phi(0) = \frac{1}{4\pi R^2}\oint_{r=R} \Phi\, da.}$$

Since we chose the center point to be the origin only for convenience, the general result is

$$\boxed{\Phi(\mathbf{r}_0) = \frac{1}{4\pi R^2}\oint_{|\mathbf{r}-\mathbf{r}_0|=R} \Phi(\mathbf{r})\, da.}$$

This proves the mean value theorem.

5. Alternative derivation using the spherically averaged potential

It is useful to give a second derivation that is very physical.

Define the spherical average of the potential about the point $\mathbf{r}_0 = 0$ by

$$\overline{\Phi}(R) = \frac{1}{4\pi}\int \Phi(R,\theta,\phi)\, d\Omega.$$

We will show directly that $\overline{\Phi}(R)$ is independent of R in charge-free space. If that is true, then

$$\overline{\Phi}(R) = \lim_{R\to 0} \overline{\Phi}(R) = \Phi(0),$$

which is exactly the theorem.

Differentiate with respect to R:

$$\frac{\mathrm{d}\overline{\Phi}}{\mathrm{d}R} = \frac{1}{4\pi}\int \frac{\partial \Phi}{\partial R}\, d\Omega.$$

Multiply by R^2:

$$R^2 \frac{\mathrm{d}\overline{\Phi}}{\mathrm{d}R} = \frac{1}{4\pi}\int R^2 \frac{\partial \Phi}{\partial R}\, d\Omega.$$

Since

$$da = R^2\, d\Omega,$$

this becomes

$$R^2 \frac{\mathrm{d}\overline{\Phi}}{\mathrm{d}R} = \frac{1}{4\pi}\oint_{r=R} \frac{\partial \Phi}{\partial r}\, da.$$

Now use Gauss's theorem:

$$\oint_{r=R} \frac{\partial \Phi}{\partial r}\, da = \int_{r<R} \nabla^2 \Phi\, d\tau.$$

But in charge-free space,

$$\nabla^2\Phi = 0.$$

Therefore

$$\oint_{r=R} \frac{\partial\Phi}{\partial r}\, da = 0.$$

Hence

$$R^2 \frac{\mathrm{d}\overline{\Phi}}{\mathrm{d}R} = 0,$$

so

$$\frac{\mathrm{d}\overline{\Phi}}{\mathrm{d}R} = 0.$$

Thus the average $\overline{\Phi}(R)$ is independent of R.

Take the limit as $R \to 0$. Since Φ is continuous,

$$\lim_{R\to 0} \overline{\Phi}(R) = \Phi(0).$$

Therefore for every R,

$$\overline{\Phi}(R) = \Phi(0),$$

that is,

$$\boxed{\Phi(0) = \frac{1}{4\pi}\int \Phi(R,\theta,\phi)\, d\Omega}$$

or

$$\boxed{\Phi(0) = \frac{1}{4\pi R^2}\oint_{r=R} \Phi\, da.}$$

This is the same result, obtained in a very direct way from Gauss's theorem and Laplace's equation.

6. Physical interpretation

The theorem says that in empty space the potential at a point is completely balanced by the values of the potential on any surrounding sphere. This is one reason harmonic functions cannot have isolated maxima or minima in charge-free regions: if the value at the center were larger than all surrounding values, it could not equal their average.

In electrostatics, this reflects the absence of sources inside the sphere. If there is no charge inside, then the field lines neither begin nor end there, and the potential is determined in a smooth, source-free way by the boundary values around the point.

Final result

For any sphere lying entirely in a charge-free region,

$$\boxed{\nabla^2\Phi = 0 \quad\Longrightarrow\quad \Phi(\mathbf{r}_0) = \frac{1}{4\pi R^2}\oint_{|\mathbf{r}-\mathbf{r}_0|=R} \Phi(\mathbf{r})\, da.}$$

That is, the electrostatic potential at the center equals the average of the potential over the spherical surface.

Problem 1.15

20.1 Problem Statement

Prove Thomson's theorem: If a number of surfaces are fixed in position and a given total charge is placed on each surface, then the electrostatic energy in the region bounded by the surfaces is an absolute minimum when the charges are placed so that every surface is an equipotential, as happens when they are conductors.

20.2 Solution

We are given a set of fixed closed surfaces

$$S_1, S_2, \ldots, S_N,$$

and on each surface S_i the total charge Q_i is prescribed. We must show that, among all possible ways of distributing those charges over the surfaces, the electrostatic energy is minimized by the special distribution for which each surface is an equipotential.

This is Thomson's theorem.

1. Strategy of the proof

We compare two charge distributions on the same set of surfaces:

- the *equilibrium* distribution, for which each surface is an equipotential;
- an arbitrary other distribution with the same total charge Q_i on each surface.

Let

$$\Phi(\mathbf{x}), \qquad \mathbf{E}(\mathbf{x}) = -\nabla\Phi$$

denote the potential and field for the equilibrium distribution, and let

$$\widetilde{\Phi}(\mathbf{x}), \qquad \widetilde{\mathbf{E}}(\mathbf{x}) = -\nabla\widetilde{\Phi}$$

denote the potential and field for any other admissible charge distribution with the same total charges on the surfaces.

We will prove that the corresponding energies satisfy

$$\widetilde{W} \geq W.$$

2. Electrostatic energy

The electrostatic energy in vacuum is

$$W = \frac{\varepsilon_0}{2}\int_V E^2\, d\tau,$$

where V is the field region outside the charged surfaces and $d\tau$ is the volume element.

So for the two configurations we have

$$W = \frac{\varepsilon_0}{2}\int_V |\mathbf{E}|^2\, d\tau, \qquad \widetilde{W} = \frac{\varepsilon_0}{2}\int_V |\widetilde{\mathbf{E}}|^2\, d\tau.$$

Define the difference field

$$\mathbf{E}_d = \widetilde{\mathbf{E}} - \mathbf{E}.$$

Then

$$\widetilde{\mathbf{E}} = \mathbf{E} + \mathbf{E}_d.$$

Substitute into $\widetilde{W}$:

$$\widetilde{W} = \frac{\varepsilon_0}{2}\int_V |\mathbf{E} + \mathbf{E}_d|^2\, d\tau.$$

Expand the square:

$$\widetilde{W} = \frac{\varepsilon_0}{2}\int_V \left(E^2 + 2\mathbf{E}\cdot\mathbf{E}_d + E_d^2\right) d\tau.$$

Thus

$$\widetilde{W} = W + \varepsilon_0\int_V \mathbf{E}\cdot\mathbf{E}_d\, d\tau + \frac{\varepsilon_0}{2}\int_V E_d^2\, d\tau.$$

So the entire proof reduces to showing that the cross term vanishes:

$$\int_V \mathbf{E}\cdot\mathbf{E}_d\, d\tau = 0.$$

Then we would obtain

$$\widetilde{W} = W + \frac{\varepsilon_0}{2}\int_V E_d^2\, d\tau \geq W,$$

which is exactly the theorem.

So now we prove that the cross term is zero.

3. The cross term

Since

$$\mathbf{E} = -\nabla\Phi, \qquad \mathbf{E}_d = -(\nabla\widetilde{\Phi} - \nabla\Phi) = -\nabla\chi,$$

where we define

$$\chi \equiv \widetilde{\Phi} - \Phi,$$

we have

$$\mathbf{E}\cdot\mathbf{E}_d = (-\nabla\Phi)\cdot(-\nabla\chi) = \nabla\Phi\cdot\nabla\chi.$$

Therefore

$$\int_V \mathbf{E}\cdot\mathbf{E}_d\,d\tau = \int_V \nabla\Phi\cdot\nabla\chi\,d\tau.$$

Now use Green's first identity:

$$\int_V \nabla\Phi\cdot\nabla\chi\,d\tau = \oint_S \chi\,\frac{\partial\Phi}{\partial n}\,da - \int_V \chi\,\nabla^2\Phi\,d\tau,$$

where $S = \bigcup_i S_i$ is the union of all the surfaces, and $\partial/\partial n$ is the derivative along the outward normal from the field region.

In the field region there is no volume charge, so the equilibrium potential satisfies Laplace's equation:

$$\nabla^2\Phi = 0.$$

Hence the volume term vanishes, and we get

$$\int_V \mathbf{E}\cdot\mathbf{E}_d\,d\tau = \oint_S \chi\,\frac{\partial\Phi}{\partial n}\,da = \sum_{i=1}^{N}\oint_{S_i} \chi\,\frac{\partial\Phi}{\partial n}\,da.$$

Now comes the key point: in the equilibrium configuration, each surface S_i is an equipotential, so

$$\Phi = V_i \qquad \text{on } S_i,$$

where V_i is a constant on that surface.

Although χ need not be constant on S_i, it is more convenient to integrate by parts the other way:

$$\int_V \nabla\Phi\cdot\nabla\chi\,d\tau = \oint_S \Phi\,\frac{\partial\chi}{\partial n}\,da - \int_V \Phi\,\nabla^2\chi\,d\tau.$$

Now both $\widetilde{\Phi}$ and Φ are harmonic in the field region, so

$$\nabla^2\widetilde{\Phi} = 0, \qquad \nabla^2\Phi = 0,$$

and therefore

$$\nabla^2\chi = 0.$$

Thus

$$\int_V \mathbf{E}\cdot\mathbf{E}_d\,d\tau = \sum_{i=1}^{N}\oint_{S_i} \Phi\,\frac{\partial\chi}{\partial n}\,da = \sum_{i=1}^{N} V_i\oint_{S_i}\frac{\partial\chi}{\partial n}\,da.$$

Now

$$\chi = \widetilde{\Phi} - \Phi,$$

so

$$\oint_{S_i}\frac{\partial\chi}{\partial n}\,da = \oint_{S_i}\frac{\partial\widetilde{\Phi}}{\partial n}\,da - \oint_{S_i}\frac{\partial\Phi}{\partial n}\,da.$$

But the total charge on surface S_i is

$$Q_i = -\varepsilon_0\oint_{S_i}\frac{\partial\Phi}{\partial n}\,da,$$

and similarly for the alternative distribution,

$$\widetilde{Q}_i = -\varepsilon_0\oint_{S_i}\frac{\partial\widetilde{\Phi}}{\partial n}\,da.$$

By hypothesis, the total charge on each surface is the same in the two configurations:

$$\widetilde{Q}_i = Q_i.$$

Therefore

$$\oint_{S_i} \frac{\partial \widetilde{\Phi}}{\partial n}\, da = \oint_{S_i} \frac{\partial \Phi}{\partial n}\, da,$$

and hence

$$\oint_{S_i} \frac{\partial \chi}{\partial n}\, da = 0.$$

So every term in the sum vanishes:

$$V_i \oint_{S_i} \frac{\partial \chi}{\partial n}\, da = 0.$$

Therefore

$$\boxed{\int_V \mathbf{E} \cdot \mathbf{E}_d \, d\tau = 0.}$$

4. Conclude the proof

Substituting this result into the energy difference,

$$\widetilde{W} = W + \frac{\varepsilon_0}{2} \int_V E_d^2 \, d\tau.$$

Since

$$E_d^2 \geq 0$$

everywhere,

$$\frac{\varepsilon_0}{2} \int_V E_d^2 \, d\tau \geq 0.$$

Hence

$$\boxed{\widetilde{W} \geq W.}$$

Thus the equilibrium distribution, the one for which each surface is an equipotential, gives the absolute minimum electrostatic energy among all charge distributions with the same total charge on each surface.

This proves Thomson's theorem.

5. When does equality occur?

Equality requires

$$\int_V E_d^2 \, d\tau = 0.$$

Since the integrand is nonnegative, this can happen only if

$$\mathbf{E}_d = 0 \qquad \text{everywhere in } V.$$

That means

$$\widetilde{\mathbf{E}} = \mathbf{E}$$

everywhere, so the two electrostatic configurations are actually the same.

Therefore the minimum is strict unless the alternative distribution coincides with the equilibrium distribution.

6. Physical interpretation

The theorem says that if conductors are allowed to redistribute charge while keeping the total charge on each conductor fixed, they do so in exactly the way that minimizes field energy.

Why? Because if some surface were not an equipotential, charges could still move along it. That motion would lower the energy until no tangential electric field remained, which is precisely the condition that the conductor be an equipotential.

So the electrostatic equilibrium configuration is the one in which the charges have arranged themselves so efficiently that any other arrangement with the same conductor charges would produce more field energy.

Final result

If the total charge on each fixed surface is prescribed, then the electrostatic energy is minimized when each surface is an equipotential. Quantitatively, for any other admissible charge distribution,

$$\boxed{\widetilde{W} - W = \frac{\varepsilon_0}{2} \int_V |\widetilde{\mathbf{E}} - \mathbf{E}|^2 \, d\tau \geq 0.}$$

Hence

$$\boxed{W \text{ is an absolute minimum at electrostatic equilibrium.}}$$